



**Mobile-Based Product Expiration Monitoring Application for Sari-Sari Store in
Palihan, Hermosa, Bataan**

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By:

Alindogan, Lester Alex D.

Boragay, Darelle D.

Dabu, Emmanuel Dave P.

Dorog, Algie R.

Santos, Ralph John B.

Villeza, Nathaniel P.

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**APPROVAL SHEET**

This Capstone Project entitled “**Mobile-Based Product Expiration Monitoring Application for Sari-Sari Store in Palihan, Hermosa, Bataan**” proposed and submitted by Boragay Darelle D., Santos Ralph John., Alindogan Lester Alex D., Dorog Algie R., Villeza Nathaniel P. and Emmanuel Dabu P. in partial fulfillment of the requirements for the degree **Bachelor of Science in Information Technology**, has been examined and found in order and is hereby recommended for acceptance and approval for **ORAL EXAMINATION**.

CAPSTONE PROJECT PROPOSAL COMMITTEE

Mrs. Jairryl Anne De Guzman MITc
Chair

Mr. Eric Almoguerra
Member

Mr. Aries John G. Cayabyab
Member

Mr. Nomer Alviado
Adviser

Accepted in partial fulfillment for the degree of Bachelor of Science in
Information Technology this 25st of May year of our Lord 2026.

Mr. Percian Joseph C. Borja, MIT
Dean, Computer Studies
Department

Date



CERTIFICATE OF PROOFREADING COMPLETION

This is to certify that **Boragay, Darelle D., Santos, Ralph John B., Dorog, Algie R., Villeza, Nathaniel P., Alindogan, Lester Alex D., and Dabu Emmanuel Dave P.** have successfully accomplished the Proofreading Phase for the Degree of Bachelor of Science in Information Technology held on **June 4, 2025**, with the proposed title, “**Mobile-Based Product Expiration Monitoring Application for Sari-Sari Store in Palihan, Hermosa, Bataan.**”

This further certifies that the students have successfully completed the proofreading requirements of the Capstone/Thesis manuscript, ensuring proper grammar, structure, formatting, and clarity of presentation, as reviewed and approved by the assigned proofreader.

Given this **4th day of June 2025** at Eastwoods Professional College of Science and Technology, Roman Superhighway, Bagong Silang, City of Balanga, Bataan.

Certified by:

Mr. Juniel D. Jenolan
Research and Curriculum Development Coordinator

Noted by:

Mrs. Jairryl Anne De Guzman
Adviser

ABSTRACT

Product expiration monitoring is essential in small retail businesses such as sari-sari stores, where manual tracking often leads to errors, product waste, and health risks. This study developed a *Mobile-Based Product Expiration Monitoring Application for Sari-Sari Stores in Palihan, Hermosa, Bataan* to provide a faster and more efficient way of managing inventory and tracking expiration dates.

The system enables store owners to record product details, monitor expiration dates, receive notification alerts, and generate reports. It also uses Barcode technology for faster input and Firebase Firestore for real-time data storage.

The study used the Rapid Application Development (RAD) methodology and evaluated the system using the ISO/IEC 25010 model. Data were gathered through surveys and interviews with store owners and IT experts.

Results showed that the system was highly acceptable in terms of functionality, usability, and reliability. The study concludes that the application improves expiration monitoring, reduces losses, and enhances inventory management.

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

Background of the Study

In today's digital age, technology plays a vital role in improving business operations and daily life. Mobile applications, in particular, have become one of the most widely used technological tools, providing fast, convenient, and efficient solutions to various problems. Many businesses now rely on mobile applications to automate tasks, reduce human errors, and improve productivity. With the continuous advancement of mobile technology, even small businesses can adopt digital systems to enhance their operations.

Globally, the improper monitoring of product expiration dates remains a significant issue, especially among small retail businesses that rely on manual record-keeping. According to the Food and Agriculture Organization of the United Nations (FAO, 2021), a large portion of food waste worldwide is caused by poor inventory management and failure to properly monitor product shelf life. Many small stores in developing countries still depend on handwritten lists or memory, which often results in expired products being sold, leading to financial losses and health risks.

In the Philippines, improper monitoring of product expiration is also a common concern, particularly among sari-sari stores. Many store owners continue to use manual methods such as notebooks or paper-based records to track product details. According to the Department of Trade and Industry (DTI, 2022), micro-retail businesses face difficulties in adopting digital inventory systems due to limited access to technology and lack of proper training. This situation contributes to inefficiencies in inventory management and increases the risk of selling expired products.

The beneficiaries of this study are the sari-sari store owners located in Brgy. Palihan, Hermosa, Bataan. These small retail stores play a crucial role in the community by providing accessible and affordable everyday goods. They serve as a primary source of basic commodities for local residents, making their operations essential to daily living in the area.

Currently, most sari-sari store owners in Palihan rely on traditional methods of managing their inventory. These methods include writing product details and expiration dates in notebooks or keeping records on paper. In some cases, store owners depend on memory to monitor product shelf life, which can be unreliable and inefficient.

Due to the reliance on manual processes, store owners often face several challenges. These include difficulty in tracking product expiration dates, lack of timely reminders, and increased chances of human error. As a result, expired products may not be identified immediately, leading to financial losses and potential health risks for consumers. Additionally, manual monitoring consumes time and effort, reducing overall efficiency in store management.

To address these challenges, the researchers proposed the development of a Mobile-Based Product Expiration Monitoring Application. This system aims to replace traditional paper-based methods with a digital and user-friendly platform. The application will enable store owners to record product information, track expiration dates efficiently, and receive automated notifications before products expire. Through this solution, the study seeks to improve inventory management, reduce product waste, and enhance the overall productivity of sari-sari store owners.

Statement of the Problem

This study aims to identify the key challenges and requirements involved in developing a Mobile-Based Product Expiration Monitoring Application for sari-sari stores in Palihan, Hermosa, Bataan. By understanding the needs of store owners, the researchers can design a system that efficiently addresses manual tracking limitations, ensures accurate monitoring of product expiration dates, and improves overall inventory management.

1. What features and functionalities should be implemented in the Mobile-Based Product Expiration Monitoring Application for Sari-Sari Store in Palihan, Hermosa, Bataan?
2. What tools and technologies could be integrated into the Mobile-Based Product Expiration Monitoring Application for Sari-Sari Store in Palihan, Hermosa, Bataan?
3. How would the system be evaluated and tested based on the following standards:
 - a. Functional Suitability
 - b. Performance Efficiency
 - c. Compatibility
 - d. Interaction Capability
 - e. Reliability
 - f. Security
 - g. Maintainability
 - h. Portability

Objectives of the Study

The primary objective of this study is to develop a functional and user-friendly Mobile-Based Product Expiration Monitoring Application that meets the needs of sari-sari store owners. Specifically, the study seeks to identify the system's essential features and functionalities, determine the appropriate tools and technologies for its development, and evaluate its performance, usability, and reliability against recognized standards. The objective of the study are the following:

1. To determine the features and functionalities to be implemented in the Mobile-Based Product Expiration Monitoring Application for the Sari-Sari Store in Palihan, Hermosa, Bataan
2. To identify the tools and technologies to be integrated for developing a Mobile-Based Product Expiration Monitoring Application for Sari-Sari Store in Palihan, Hermosa, Bataan.
3. To evaluate and test the system's functionality based on the ISO/IEC 25010.

Significance of the Study

The significance of this study refers to the importance and potential impact of developing a Mobile-Based Product Expiration Monitoring Application for sari-sari stores. By automating the tracking of product expiration dates, the system aims to improve inventory management, prevent financial losses, and ensure consumer safety.

Sari-Sari Store Owners – The primary beneficiaries of this study are the sari-sari store owners, who play a crucial role in supplying everyday goods to the local community. The application will provide them with an efficient way to monitor product expiration dates, reduce the risk of selling expired items, and improve overall inventory management. By using the system, store owners can save time, reduce errors associated

with manual notebook tracking, and make informed decisions regarding stock replenishment and product rotation.

Future Researchers – The study serves as a reference for future researchers who wish to explore mobile-based inventory management solutions, offering insights into practical system development, implementation, and evaluation in small-scale retail contexts.

Scope and Delimitation

The primary focus of this research is to design and develop a Mobile-Based Product Expiration Monitoring Application for Sari-Sari Stores in Palihan, Hermosa, Bataan. The system is intended to help store owners efficiently monitor product expiration dates, minimize product waste, and prevent the sale of expired goods. The application will be implemented in selected sari-sari stores in Palihan, Hermosa, Bataan, to provide convenience for store owners and consumers.

The researchers establish the following parameters:

Store Owner (User):

1. Add, edit, and delete product information.
2. Set and update product expiration dates.
3. Monitor near-expiry and expired products.
4. Receive notifications and alerts for expiring products.
5. Generate expiration reports.
6. Manage inventory records.
7. Use barcode or automatic entry for faster and more accurate product input.
8. Access cloud-based data storage for secure backup and to enable device switching without data loss.

9. Track product movements in and out of inventory, including recording low stock, losses, or damages.
10. Automatically update inventory when products are scanned out, reflecting real-time stock changes.
11. Separate expiration tracking from general inventory for better monitoring and management of expiring products.

Administrator (Admin):

1. Review and manage store owner registrations.
2. Approve or reject store owner accounts.
3. View all registered and approved stores.
4. Manage disabled store accounts.
5. Update and verify business permit details of stores.
6. Re-enable stores with expired business permits after validation.
7. Review and manage store requests such as profile updates, email changes mobile number changes, and store closure requests.
8. View application data, ratings, and feedback submitted by users.
9. Monitor notifications and alerts related to store approvals, requests, and system updates.

Definition of Terms

Barcode Technology – A system that uses machine-readable codes to quickly identify and input product information into the application.

Cloud Storage – An online service that allows data to be stored, accessed, and backed up over the internet.

Digitalization – The process of converting manual store operations into a computerized system using digital tools.

Expiration Monitoring – The process of tracking product expiration dates to prevent the sale of expired goods.

Inventory Management – The process of tracking, organizing, and controlling product stocks within a store.

Mobile Application – A software program designed to run on mobile devices such as smartphones and tablets.

Notification Alert – A system-generated message that notifies the user about important events, such as products nearing expiration.

Product Expiration – The date after which a product is no longer safe or suitable for use or consumption.

Sari-Sari Store – A small neighborhood retail store in the Philippines that sells basic goods and daily necessities.

Automation – The use of technology to perform tasks with minimal human intervention, reducing manual effort and increasing efficiency.

Consumer Safety – The protection of buyers from harmful or unsafe products, particularly those that are expired or contaminated.

Digital Platform – A technology-based environment (such as a mobile application) that allows users to perform tasks like data input, monitoring, and communication.

Digital Transformation – The process of integrating digital technology into business operations to improve performance and service delivery.

Food Waste – The loss or disposal of food that is still suitable for consumption, often caused by poor inventory management or expiration neglect.

Human Error – Mistakes made by individuals during manual processes, such as incorrect recording or failure to monitor expiration dates.

Inefficiency – The inability to achieve maximum productivity due to wasted time, effort, or resources.

Inventory Visibility – The ability to accurately monitor and track stock levels, product movement, and availability in real time.

Manual Record-Keeping – The traditional method of recording data using paper, notebooks, or memory instead of digital systems.

Micro-Retail Business – A small-scale retail operation, such as a sari-sari store, that sells basic goods to a local community.

Mobile Technology – Portable digital devices (such as smartphones and tablets) used to run applications and perform various tasks.

Operational Productivity – The efficiency with which a business completes tasks and manages resources to achieve better output.



Product Shelf Life – The length of time a product remains usable, safe, or fit for consumption before it expires.

Real-Time Processing – The immediate updating and processing of data as soon as it is entered into the system.

Technological Adoption – The process of accepting and using new technologies in daily operations or business activities.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter presents related literature and studies that provide background information and support for the development of the Mobile-Based Product Expiration Monitoring Application for Sari-Sari Stores in Palihan, Hermosa, Bataan. The researchers reviewed concepts, systems, technologies, and existing studies related to mobile-based inventory systems, product expiration monitoring, sari-sari store digitalization

And notification and alert Systems. Related materials were gathered from journals, theses, online articles, and government publications.

Mobile-Based Inventory Systems

According to Hernandez and Smith (2025), mobile-based inventory systems enhance inventory visibility, allowing business owners to monitor stock levels in real time and make informed decisions. Their findings highlight the importance of accessibility and transparency in improving overall stock management efficiency.

Similarly, Garcia and Reyes (2024) found that the implementation of mobile inventory systems resulted in a 25% increase in operational efficiency among small businesses. This suggests that digital tools significantly reduce delays and improve workflow.

Furthermore, Lopez et al. (2023) stated that mobile platforms provide real-time updates, enabling faster decision-making and reducing product loss. This emphasizes that timely data access is critical in managing inventory effectively.

Meanwhile, Reyes and Dela Cruz (2022) emphasized that mobile inventory solutions are ideal for micro-enterprises due to their affordability and ease of use.

Their study revealed that store owners who adopted such systems improved stock accuracy by more than 40%.

In addition, Singh and Patel (2021) noted that mobile inventory applications allow business owners to efficiently track stock levels, product movement, and inventory status using smartphones. As a result, human error associated with manual record-keeping is significantly reduced.

Product Expiration Monitoring

According to Adams and White (2025), predictive analytics can be integrated into expiration monitoring systems to forecast expiration patterns and reduce product waste. This highlights the future potential of intelligent inventory systems.

Similarly, Davis and Brown (2024) emphasized the importance of integrating expiration tracking with inventory management systems to improve efficiency and reduce losses caused by expired products.

Moreover, Chen and Wang (2023) found that real-time expiration alerts can reduce food waste by up to 50%, demonstrating the effectiveness of automated alert systems in minimizing losses.

In line with this, Park and Kim (2022) concluded that automated expiration monitoring systems significantly increase product turnover and reduce health risks for consumers.

However, the Food and Agriculture Organization (FAO, 2021) reported that improper tracking of expiration dates remains a major cause of food waste worldwide, especially in small retail environments, indicating that many businesses still rely on inefficient methods.

Meanwhile, Gomez and Hernandez (2020) developed a digital expiration tracking system that automatically notifies store owners of products nearing expiration. Their study showed a 35% reduction in expired-product losses, supporting the effectiveness of automated monitoring systems.

Sari-Sari Store Digitalization

According to Lim and Tan (2025), peer-to-peer training programs are effective in promoting digital adoption among sari-sari store owners, particularly in communities with limited access to formal training.

Similarly, Reyes and Garcia (2024) indicated that simplified mobile applications can overcome technical barriers faced by store owners, making digital tools more accessible and user-friendly.

Furthermore, Cruz and Santos (2023) observed that digital literacy programs significantly increase the adoption of inventory systems among small retailers by improving users' technical skills.

However, the Department of Trade and Industry (DTI, 2022) reported that only a small percentage of sari-sari stores use digital tools, highlighting the need for more technological interventions in the sector.

In contrast, Torres and Medina (2021) found that although lack of technical knowledge and access to digital tools are major barriers to digital transformation, adoption still leads to improved sales monitoring and reduced losses.

Notification and Alert Systems

According to Garcia and Hernandez (2025), multi-channel notification systems, such as SMS and email alerts, improve the delivery rate and effectiveness of alerts in inventory systems.

Meanwhile, Brown and Davis (2025) suggested that integrating machine learning can optimize notification timing, ensuring that alerts are delivered when users are most likely to respond.

Furthermore, Kim and Park (2024) demonstrated that personalized notifications enhance user engagement and improve response rates in monitoring systems.

Similarly, Santos and Lim (2023) developed a notification-based inventory system that sends alerts for near-expiry products, resulting in a 45% improvement in response efficiency.

In addition, Chen and Lee (2022) stated that push notifications significantly improve user response time in time-sensitive management systems.

Small Retail Inventory Applications

According to Hernandez and Mendoza (2025), inventory applications help reduce stockouts and overstocking, thereby improving overall business performance for small retailers.

Similarly, Santos and Garcia (2025) suggested that integrating payment processing into inventory systems streamlines both sales and inventory management.

Moreover, Reyes and Cruz (2024) emphasized that user-friendly interfaces are crucial for the adoption of inventory applications among small retail store owners.

Meanwhile, Tan and Lim (2023) found that cloud-based inventory systems offer scalability, making them suitable for growing businesses.



In addition, Mendoza (2022) developed a mobile inventory application for mini-groceries that improved stock accuracy and reduced expired goods, emphasizing the importance of simplicity and offline capability.

Finally, Wu et al. (2021) stated that lightweight inventory systems designed for small retail shops provide essential features such as stock tracking, expiration monitoring, and reporting.

CHAPTER III

RESEARCH DESIGN AND METHODOLOGY

This chapter presents related literature and studies that provide background information and support for the development of the Mobile-Based Product Expiration Monitoring Application, focusing on key concepts such as mobile inventory system, product expiration monitoring, inventory management, sari-sari store digitalization, notification alert system, and cloud-based inventory solutions. The researchers reviewed relevant systems, technologies, and existing studies related to barcode technology, real-time data processing, and small retail inventory applications. Related materials were gathered from journals, theses, online articles, and government publications.

Research Methods

The researchers used a mixed-methods approach, combining qualitative and quantitative methods to evaluate the effectiveness and usability of the proposed Mobile-Based Product Expiration Monitoring Application. Qualitative methods were used to understand the experiences, needs, and challenges of sari-sari store owners in manually managing product expiration. In contrast, quantitative methods were applied to measure system usability, efficiency, and user satisfaction.

An interview method and survey method were conducted to assess the system's performance in terms of ease of use, accuracy in monitoring expiration dates, notification effectiveness, and overall user satisfaction. This approach allowed the researchers to obtain both experiential insights and numerical data necessary for evaluating the system.

Research Locale

This study was conducted in Palihan, Hermosa, Bataan, Residence, IT Experts and Admin served as the primary respondents. The locale also represents a typical small-scale retail environment where digital solutions can significantly improve daily store operations.

Respondents of the Study

This section details the respondents who participated in the evaluation and validation of the proposed system. The respondents were purposively selected based on their knowledge, experience, and relevance to the objectives of the study. The respondents included store owners and clients from Palihan, Hermosa, Bataan.

A total of (110) respondents participated in this research. Specifically, Ninety 99 residence, 10 IT Experts and 1 Admin who completed the survey to evaluate the system's usability, functionality, and overall performance.

In 2020, Brgy. Palihan, Hermosa, Bataan had a population of 6,556 residents.

$N = 6,556$ (Residence of Barangay Palihan)

$e = 0.1$

$$n = \frac{N}{(1 + Ne^2)}$$

$$n = \frac{6,556}{(1 + 6,556 (0.1)^2)}$$

$$n = 99.07 / 99 \text{ Residence}$$

Where:

IT Professionals = 10

Administrator = 1

Residence = 99

n = IT professionals, Administrator, Residence

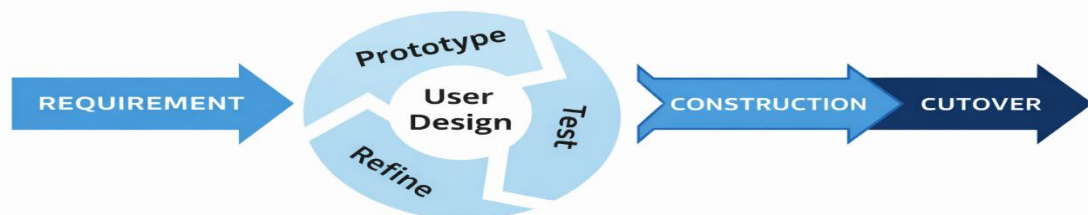
n = 110 respondents

Software Development Methodology

The researchers used the Rapid Application Development (RAD) model to develop the system. RAD is a software development methodology that emphasizes rapid prototyping, iterative development, and continuous user feedback. This model was chosen due to its flexibility, speed, and ability to incorporate user suggestions throughout the development process. Unlike traditional development models, RAD allows the researchers to deliver functional prototypes quickly and refine them based on actual user experience.

A study by Smith and Jones (2020) also employed the RAD methodology in developing a similar inventory management system, highlighting its effectiveness in delivering user-centric solutions promptly. Their findings support the suitability of RAD for projects requiring rapid iteration and user involvement.

Figure 1. Graphical Model of RAD Methodology



Development Process based on Chosen SDLC

Requirement Phase

This phase focuses on identifying the system objectives, scope, and user requirements. The researchers gathered the necessary information from sari-sari store owners regarding their current method of monitoring product expiration. The data gathered formed the foundation for the system's design and development.

User Design Phase

This phase focuses on developing the system architecture, interface design, and initial prototypes. The researchers designed the system layout and workflows based on user requirements. Feedback from the intended users was gathered and incorporated to ensure that the system is user-friendly and functional.

Construction Phase

This phase involves the system's actual development. The researchers programmed the core features of the application, including product input, expiration monitoring, notification alerts, report generation, and inventory tracking. Continuous testing and user feedback were conducted to ensure system accuracy and performance.

Cutover Phase

This is the implementation phase, in which the finalized system is prepared for actual deployment. This phase includes system testing, data conversion, user training, and system deployment. It ensures that the system is fully functional, reliable, and ready for real-world use.

Research Instruments

The researchers used the following research instruments in gathering data:

Interviews – Interviews were conducted with selected store owners in Palihan to gather information about their experiences, challenges, and expectations regarding expiration monitoring.

Online Research – The researchers also conducted online research by reviewing related literature, studies, articles, and reliable sources from the internet.

System Development Tools

The following tools were used for the design and documentation of the system:

Mockup – Visual representations of the system interface and layout.

Entity-Relationship Diagram (ERD) – Illustrates the relationship between data entities in the system.

Context Level Diagram - A high-level diagram that shows the system as a single process and how it interacts with external entities through data flows.

Use Case Diagram – A diagram that shows the interactions between the user (store owner) and the system.

Data Flow Diagram (DFD) – A diagram that illustrates how data moves within the system.

Technologies to be Used

The researchers utilized the following technologies:

Java – The primary programming language used for developing the Android mobile application. It provides a robust and reliable environment for building native Android apps.

XML – Used for designing Android layouts, defining UI components, and structuring application screens.

Emulators & Android Devices – Used to test the application across different screen sizes, Android versions, and real-world conditions.

Gradle – The build automation system used in Android Studio to manage dependencies and compile the project.

Android Studio – The integrated development environment (IDE) used to write, compile, and debug Java code for the Android application.

Cloud Storage – Used to store and back up inventory and expiration data online securely. Cloud storage enables data synchronization, prevents data loss, and allows access to records across multiple devices when needed.

Barcode Technology – Used for quick and accurate product identification. Barcodes allow store owners to easily add, update, or retrieve product information by scanning, reducing manual data entry and errors.

Kotlin – A modern programming language officially supported by Android. Kotlin is used alongside Java to improve code readability, reduce boilerplate code, and enhance overall application performance and safety.

Firebase Firestore – A flexible, scalable cloud database used to store structured inventory and expiration data with real-time updates.

Software Evaluation

The developed system undergone a comprehensive evaluation to determine its overall quality, performance, and user satisfaction. This evaluation was conducted using the ISO/IEC 25010 Software Quality Model, an internationally recognized standard for assessing software product quality. The selection of this model ensures a structured and standardized approach to evaluating the system's attributes.

The evaluation process involved both survey questionnaires and interviews. Survey questionnaires were distributed to store owners and selected Information Technology (IT) professionals to gather quantitative data on various aspects of the system. Interviews were conducted with key stakeholders to obtain qualitative insights and in-depth feedback.

CHAPTER IV

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents, analyzes, and interprets the data gathered during the development and evaluation of the Mobile-Based Product Expiration Monitoring Application for Sari-Sari Store in Palihan, Hermosa, Bataan. The purpose of this chapter is to describe how the system was designed, implemented, and tested to address the challenges faced by sari-sari store owners in monitoring product expiration dates.

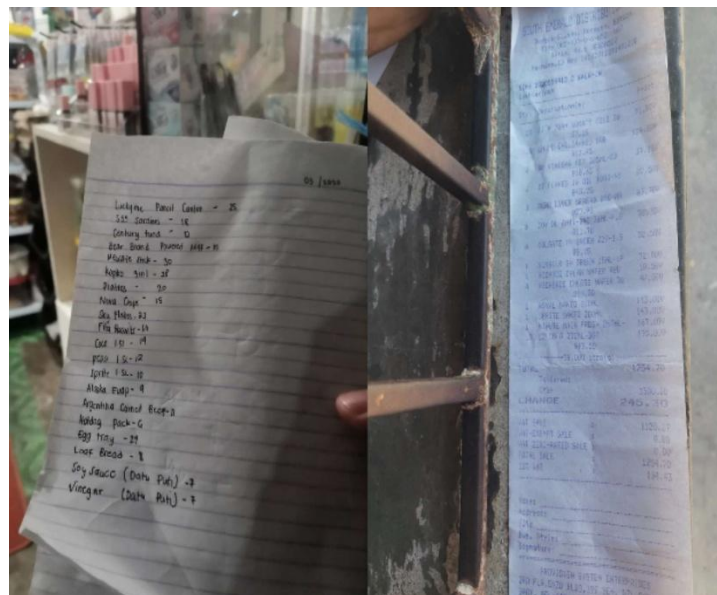
The discussion focuses on the development of the system using a user-centered approach to ensure that the application is simple, accessible, and effective for sari-sari store owners. It also highlights the features and functionalities of the system that help store owners track product expiration dates, receive notifications for expiring products, and manage inventory more efficiently.

The analysis further demonstrates how the system improves product monitoring, reduces the risk of selling expired products, and supports better inventory management for small retail businesses.

Traditional Process

Before the development of the Mobile-Based Product Expiration Monitoring Application, sari-sari store owners in Palihan, Hermosa, Bataan relied on traditional manual methods to track product expiration dates. These methods typically involved writing product details and expiration dates in notebooks, checking product labels manually, or relying on memory.

Figure 2. Traditional Process of Product List



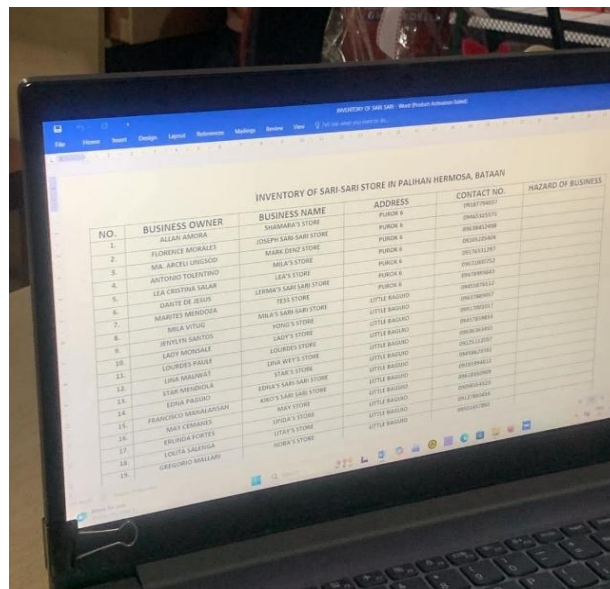
The Figure 2 shows traditional method used by store owners to monitor product list using notebook or receipt record.

Figure 3. Manual Product Checking



Figure 3 shows how store owners manually check product labels to identify expiration dates. This process is time-consuming and inefficient, especially when managing multiple products with different expiration periods.

Figure 4. Store Owner List Using Word document



NO.	BUSINESS OWNER	BUSINESS NAME	ADDRESS	CONTACT NO.	HAZARD OF BUSINESS
1.	ALLAN AMORA	SHAMARA'S STORE	PURON 8	09487790077	
2.	FLORENCE MORALES	JOSEPH SARI-SARI STORE	PURON 8	09484450088	
3.	MBA. ARCELY ORGADO	MARK DENC STORE	PURON 8	09516324046	
4.	ANTONIO YOLANTINO	MARK'S STORE	PURON 8	09516324046	
5.	ELA CRISTINA SALAM	LEA'S STORE	PURON 8	09516324046	
6.	DAVITE DE JESUS	LEONAR'S SARI-SARI STORE	PURON 8	09487790077	
7.	MARITES MENDOZA	TESS STORE	LITTLE BAGUIO	09487790077	
8.	MELA VITULO	KULA'S SARI-SARI STORE	LITTLE BAGUIO	09487790077	
9.	JENNIFER SANTOS	YONG'S STORE	LITTLE BAGUIO	09487790077	
10.	LADY MORGALL	LADY'S STORE	LITTLE BAGUIO	09487790077	
11.	LOURDES PAULE	LOURDES STORE	LITTLE BAGUIO	09487790077	
12.	LYRA MARIANET	LYRA WEE'S STORE	LITTLE BAGUIO	09487790077	
13.	LYRA MARIANET	LYRA'S STORE	LITTLE BAGUIO	09487790077	
14.	EDITH SARI-SARI STORE	EDITH'S SARI-SARI STORE	LITTLE BAGUIO	09487790077	
15.	KIM'S SARI-SARI STORE	KIM'S SARI-SARI STORE	LITTLE BAGUIO	09487790077	
16.	FRANCISCO MANGALATAN	MARK'S STORE	LITTLE BAGUIO	09487790077	
17.	MARK'S STORE	MARK'S STORE	LITTLE BAGUIO	09487790077	
18.	EDITHA SARI-SARI STORE	EDITHA'S STORE	LITTLE BAGUIO	09487790077	
19.	MARK'S STORE	MARK'S STORE	LITTLE BAGUIO	09487790077	

Figure 4 shows how store owners are recorded and managed using a Word list. This traditional method involves manually encoding and updating store owner information.

Development of the System

The development of the Mobile-Based Product Expiration Monitoring Application for Sari-Sari Store in Barangay Palihan, Hermosa, Bataan utilized the Rapid Application Development (RAD) methodology to ensure an efficient, iterative, and user-centered approach. The researchers initially examined the limitations of manual expiration tracking and gathered system requirements through interviews and surveys with store owners, residents, IT experts, and the system administrator. These stakeholders provided both practical and technical insights, which guided the design of a streamlined, accessible, and responsive mobile application tailored for small retail environments.

The system was developed with essential modules including product management, expiration monitoring, barcode scanning, inventory tracking, and automated notification alerts for near-expiry and expired products. A cloud-based database using Firebase Firestore was integrated to support real-time data processing, secure storage, and multi-device accessibility. Throughout the development cycle, continuous testing and validation were conducted with the participation of store owners, residents, IT experts, and the administrator, ensuring that the system meets ISO/IEC 25010 quality standards. The iterative refinement process resulted in a reliable and scalable application that significantly enhances inventory visibility, reduces product waste, and promotes consumer safety.

Requirement

As part of the project requirements, the researchers visited selected sari-sari store owners in Palihan, Hermosa, Bataan to present the proposed system and explain its objectives.

The researchers also submitted formal documents, including letter to conduct, proposal letter, and permission letter, to obtain approval for conducting the study.

Figure 5. Letter to Conduct

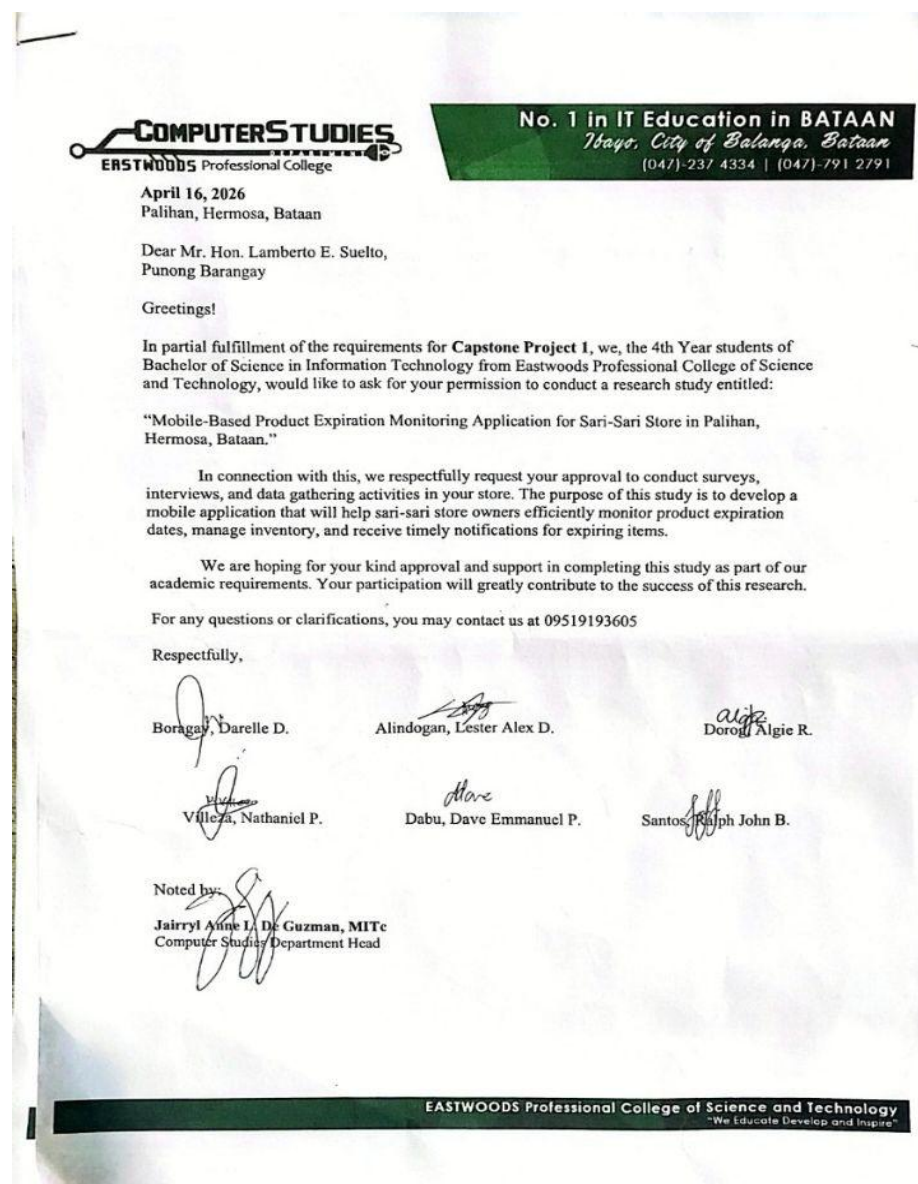


Figure 5 shows formal request for permission to carry out activities such as surveys, interviews, or testing.

Figure 6. Formal Deployment Letter

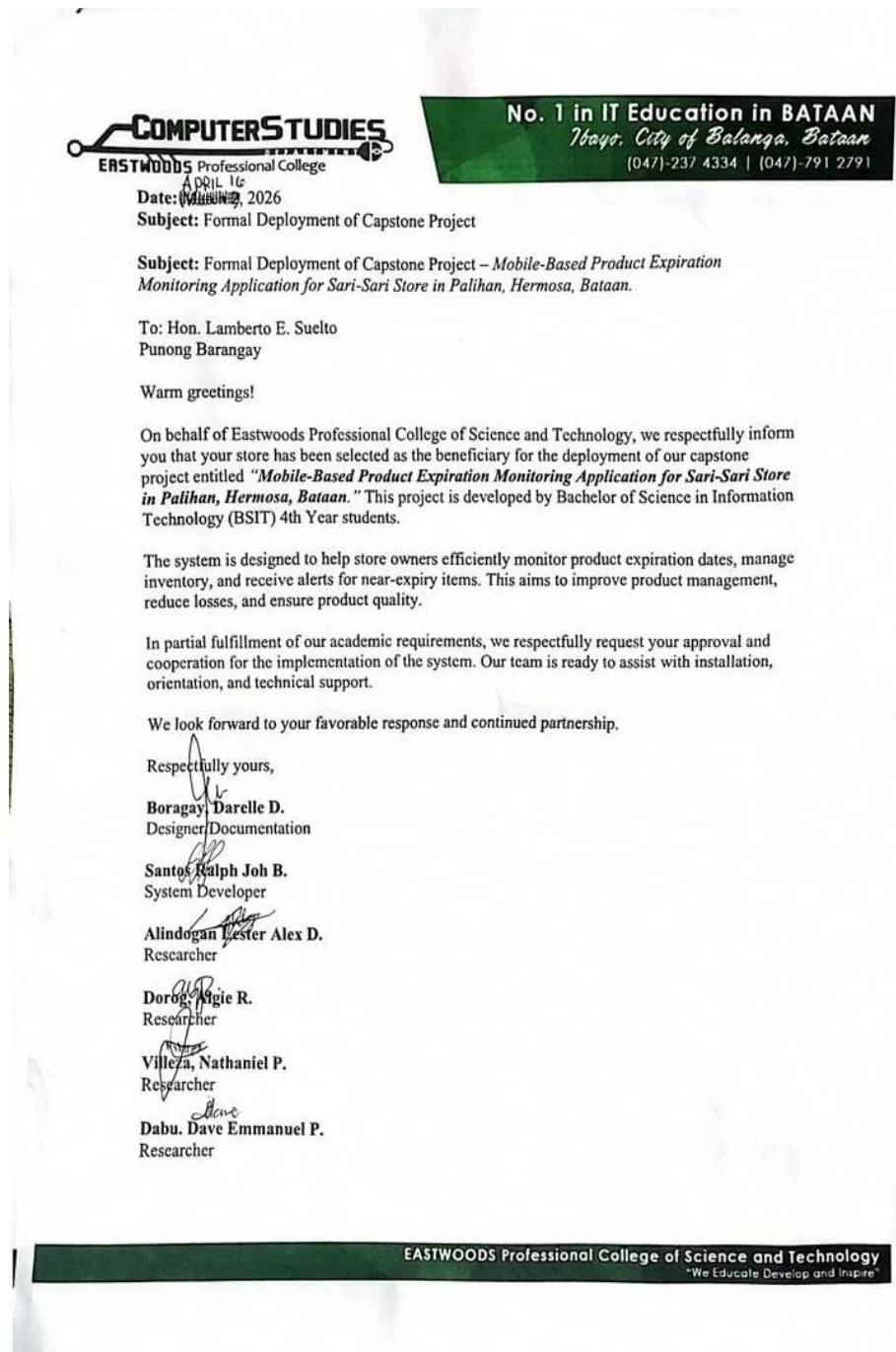


Figure 6 shows a Formal Deployment Letter, an official document informing an organization about the planned implementation of a system.

Figure 7. Proposal Letter

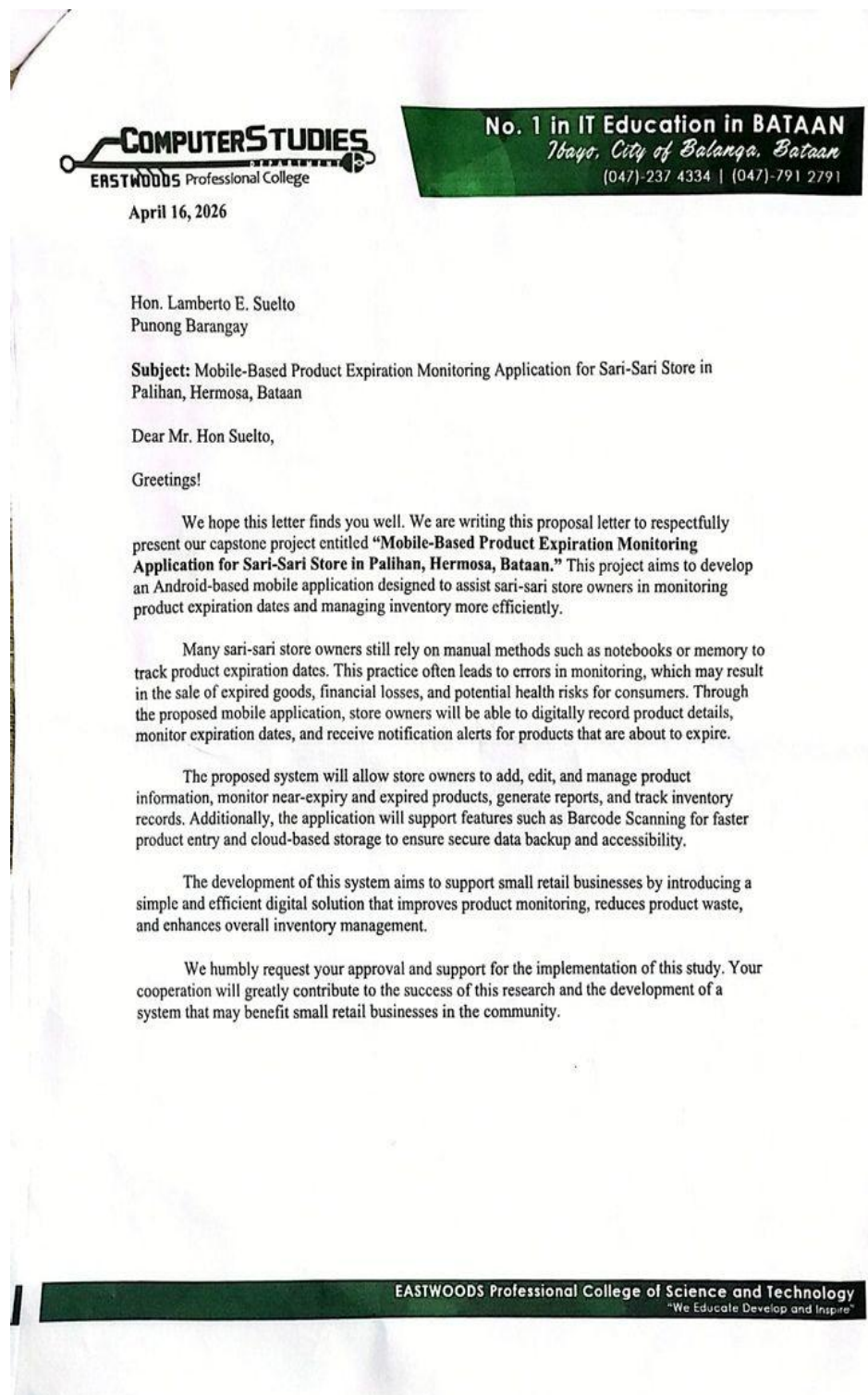


Figure 8. Proposal Letter

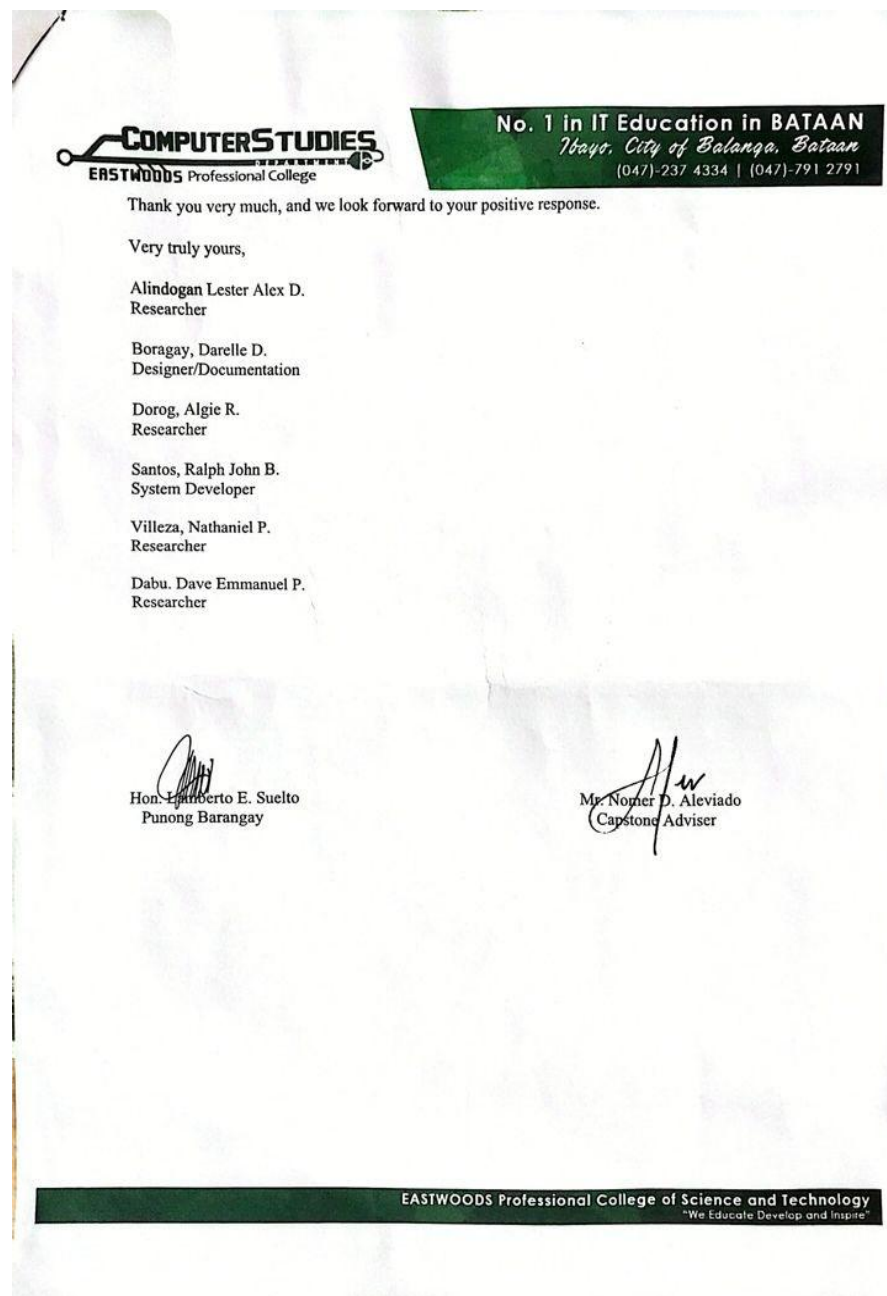


Figure 7 and 8 shows formal document presenting a project or system for approval. It outlines the objectives, scope, and expected outcomes to gain support or collaboration.

Figure 9. *Estimated Cost of the System*

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ESTIMATED COST OF THE SYSTEM

Category	Estimated Cost
Android Development Tools	P0
Firebase Firestore and Cloud Storage	P0
Internet and Data Expenses	P1,000
QR Code Scanner Integration	P500
Testing Devices & Emulators	P500
System Maintenance & Support	P1,500
Miscellaneous Expenses (printing, documentation, etc.)	P1,500
Total Estimated Cost	P5,000

EASTWOODS Professional College of Science and Technology

Figure 9 shows the estimated cost of the system, outlining the projected expenses for development, implementation, materials, and other required resources.

Figure 10. Gantt Chart

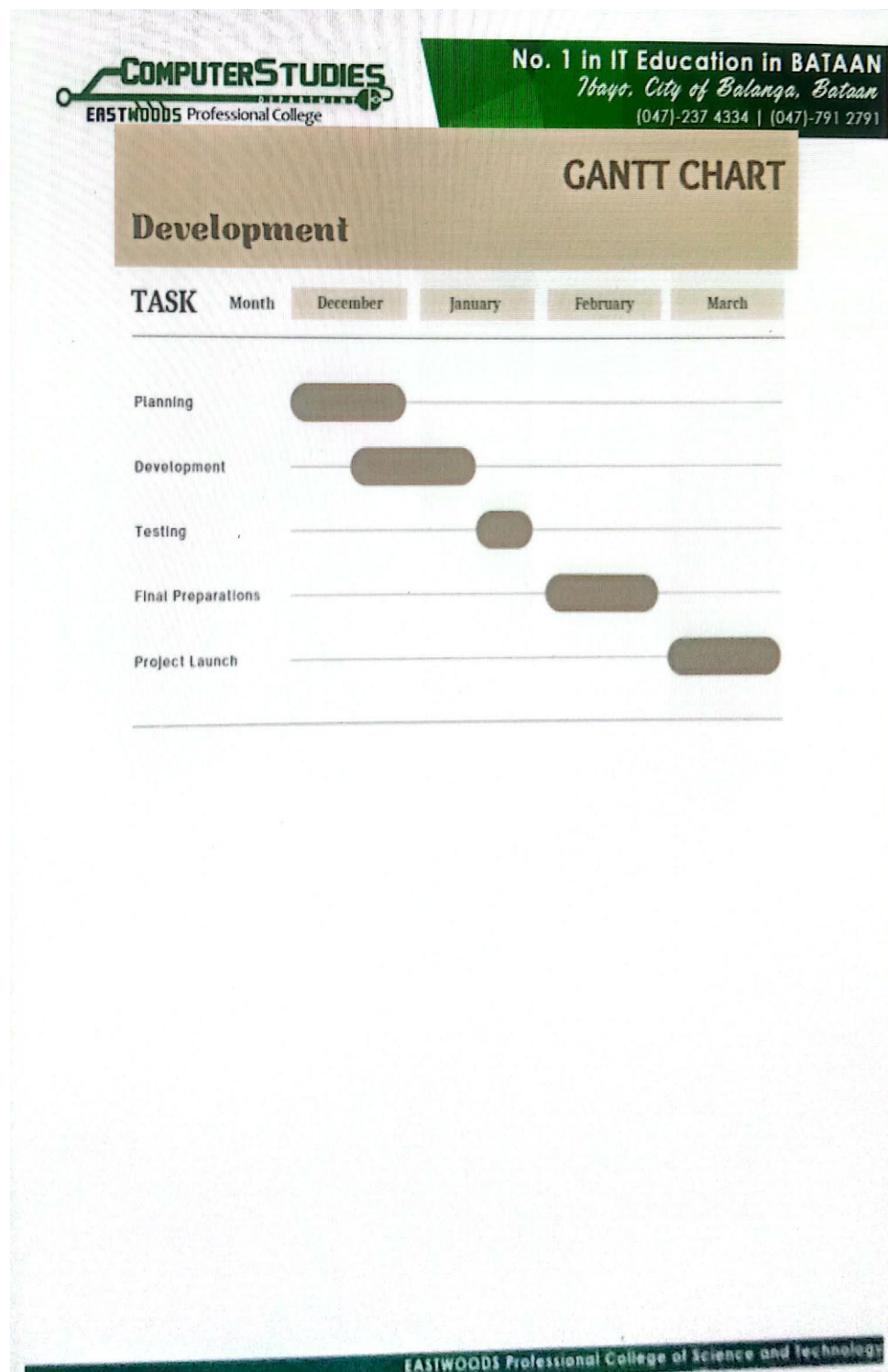


Figure 10 shows the Gantt chart, illustrating the project timeline, including task schedules, durations, and the sequence of activities for system development.

Figure 11. *Submission and Checking of Documents by Barangay Personnel*



Figure 11 shows barangay personnel receiving and checking the submitted letters and documents. This process involves verifying the completeness and correctness of the requirements before approval.

Figure 12. *Admin/Secretary of the Application*

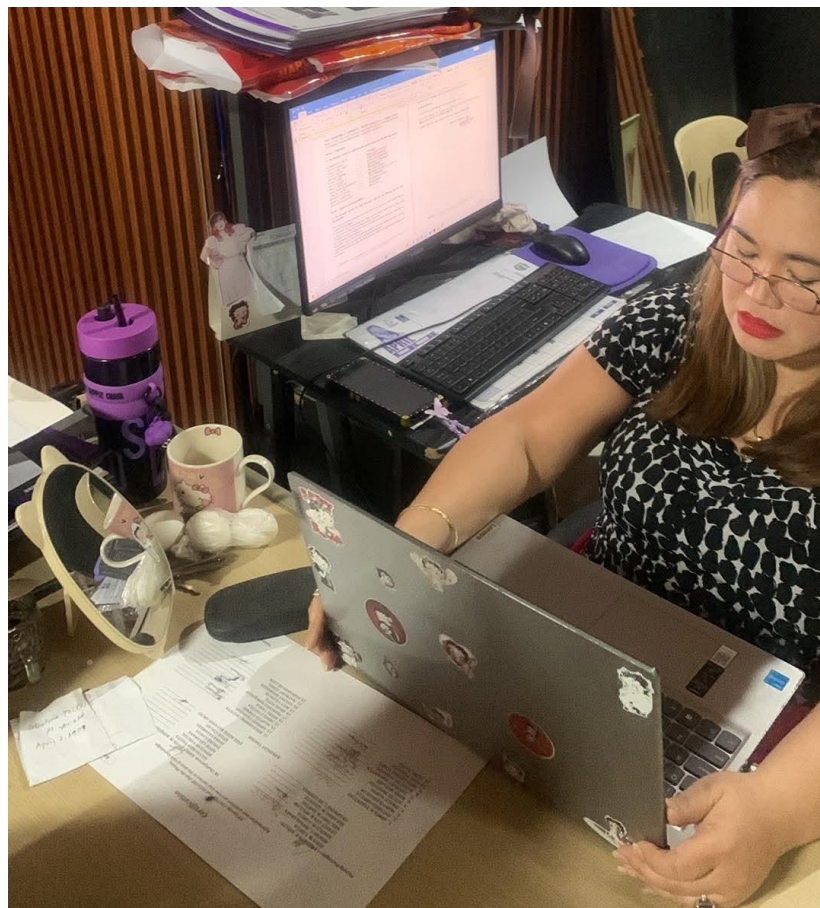


Figure 12 shows the admin or secretary of the application, where authorized users manage. This includes reviewing records, updating information, and handling user requests within the system.

Figure 13. *Conducting Interview with Store Owners*



Figure 14. *Conducting Interview with Store Owners*



Figure 13 and 14 shows the researchers conducting interview with store owners. This process was done to gather relevant data, insights, and feedback regarding their current practices and challenges in managing their stores.

Design

The process begins with the user, specifically the sari-sari store owner, who inputs data into the Inventory System. This system serves as the core of the application, where all operations such as product management, expiration tracking, and data processing take place. Through this system, the user can access different functional components including the scanner, which allows faster product entry using barcode technology; the inventory module, which manages product details and stock levels; the reports module, which generates summaries of inventory and expiration status the profile module, which handles user account settings; and the dashboard, which provides an overview of store activities and product conditions.

From the reports module, the system generates alerts based on the expiration status of products. These alerts notify the user when items are near expiration or already expired, helping prevent financial loss and ensuring consumer safety. The alerts are then sent back to the user, allowing them to take immediate action such as removing or replacing products.

All data processed within the system, including product details, expiration dates, and inventory records, are stored in data storage (save and restore). This ensures that information is securely saved, can be retrieved when needed, and remains accessible even when the user switches devices. Overall, the diagram illustrates how the system centralizes user input, processes it through various modules, generates alerts, and maintains reliable data storage, aligning with the goal of improving inventory management and expiration monitoring for sari-sari store owners.

Figure 15. Context Level Diagram

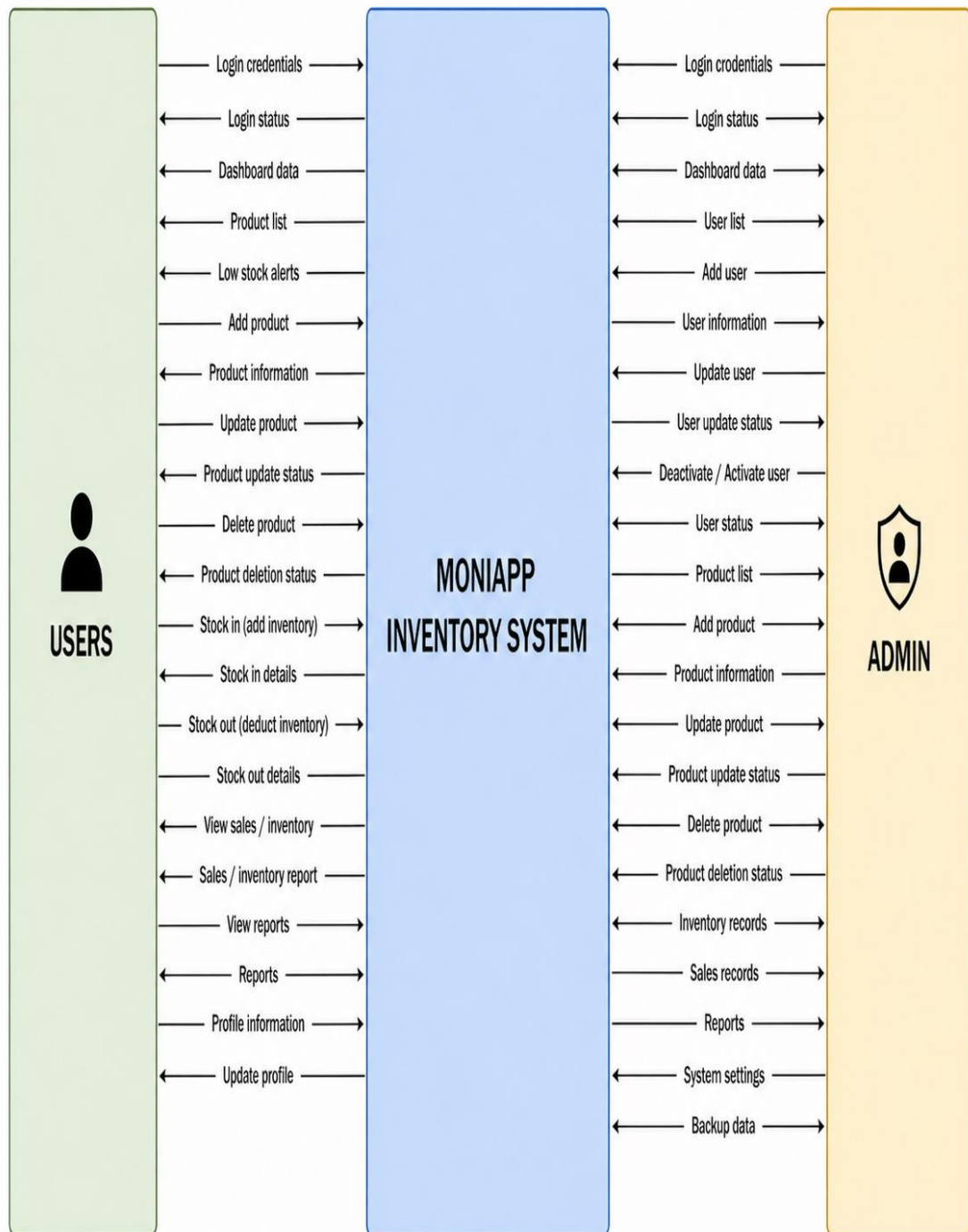


Figure 15 shows the context level diagram of the proposed system, illustrating the overall interaction between the system and its external entities

Figure 16. Entity Relationship Diagram

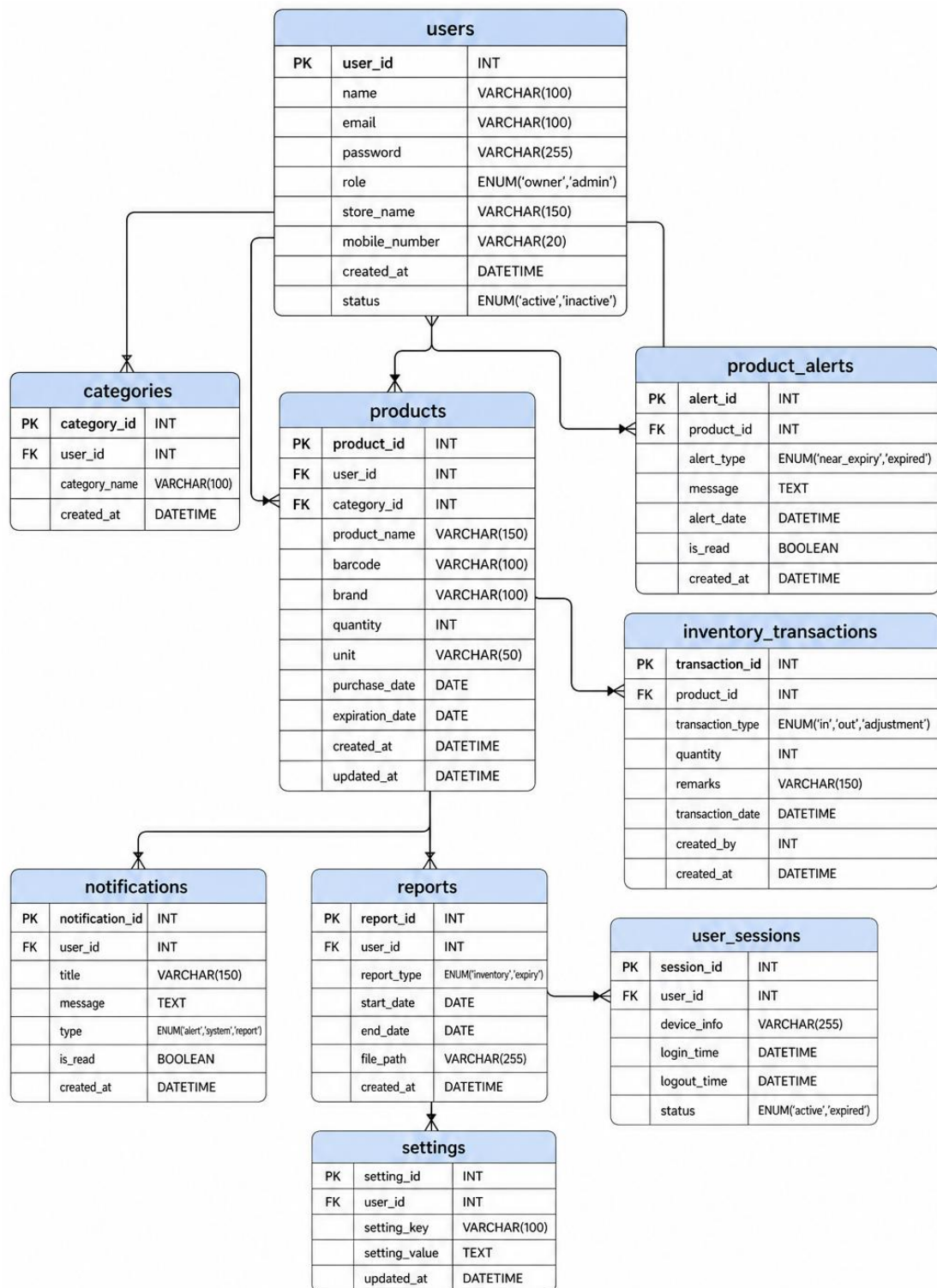


Figure 16 shows the Entity-Relationship Diagram (ERD) of the proposed system, illustrating the relationships between different entities within the database.

Figure 17. Use Case Diagram

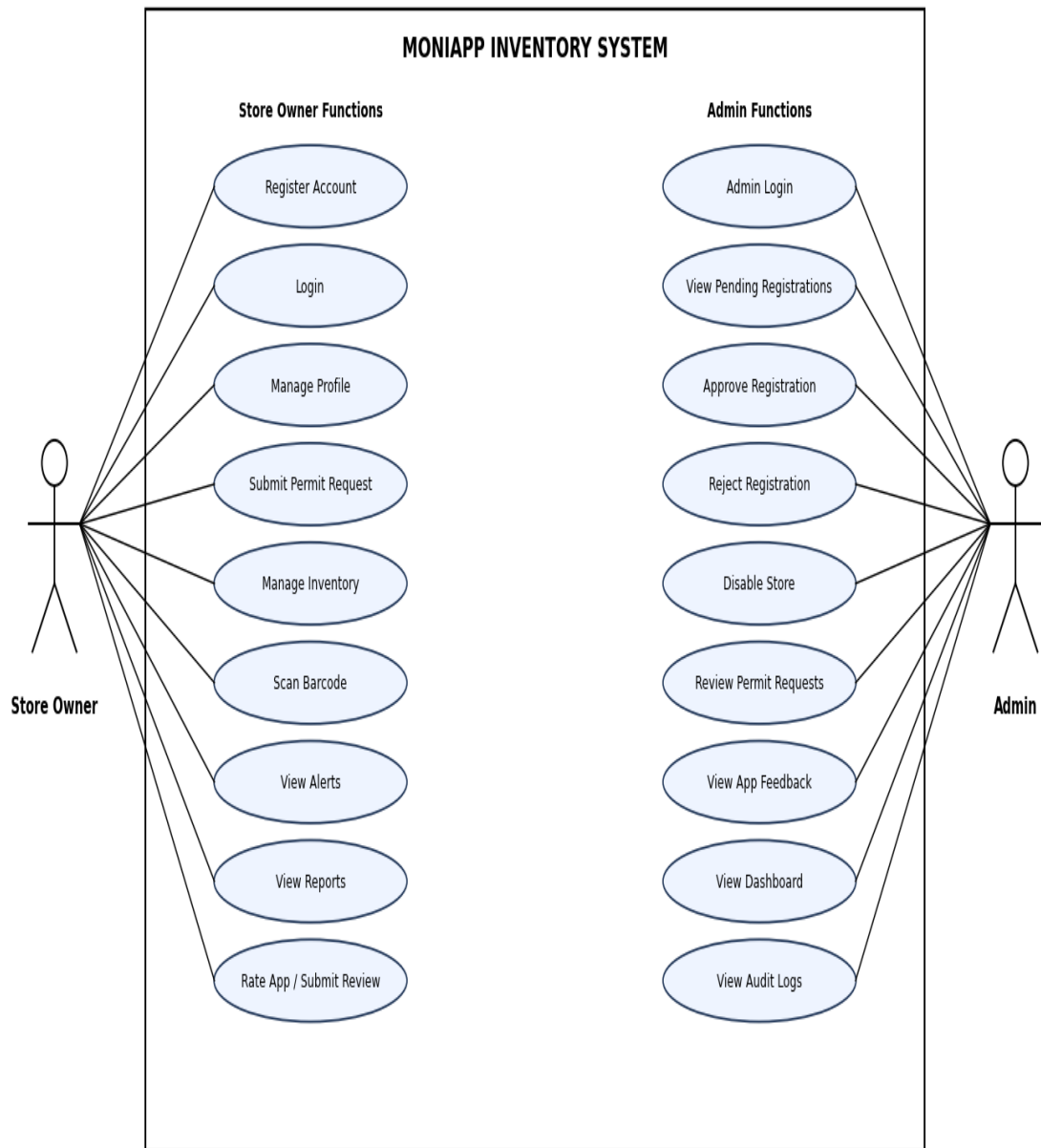


Figure 17 shows the Use Case Diagram of the proposed system, illustrating the interactions between the users (actors) and the system.

Figure 18. Data Flow Diagram

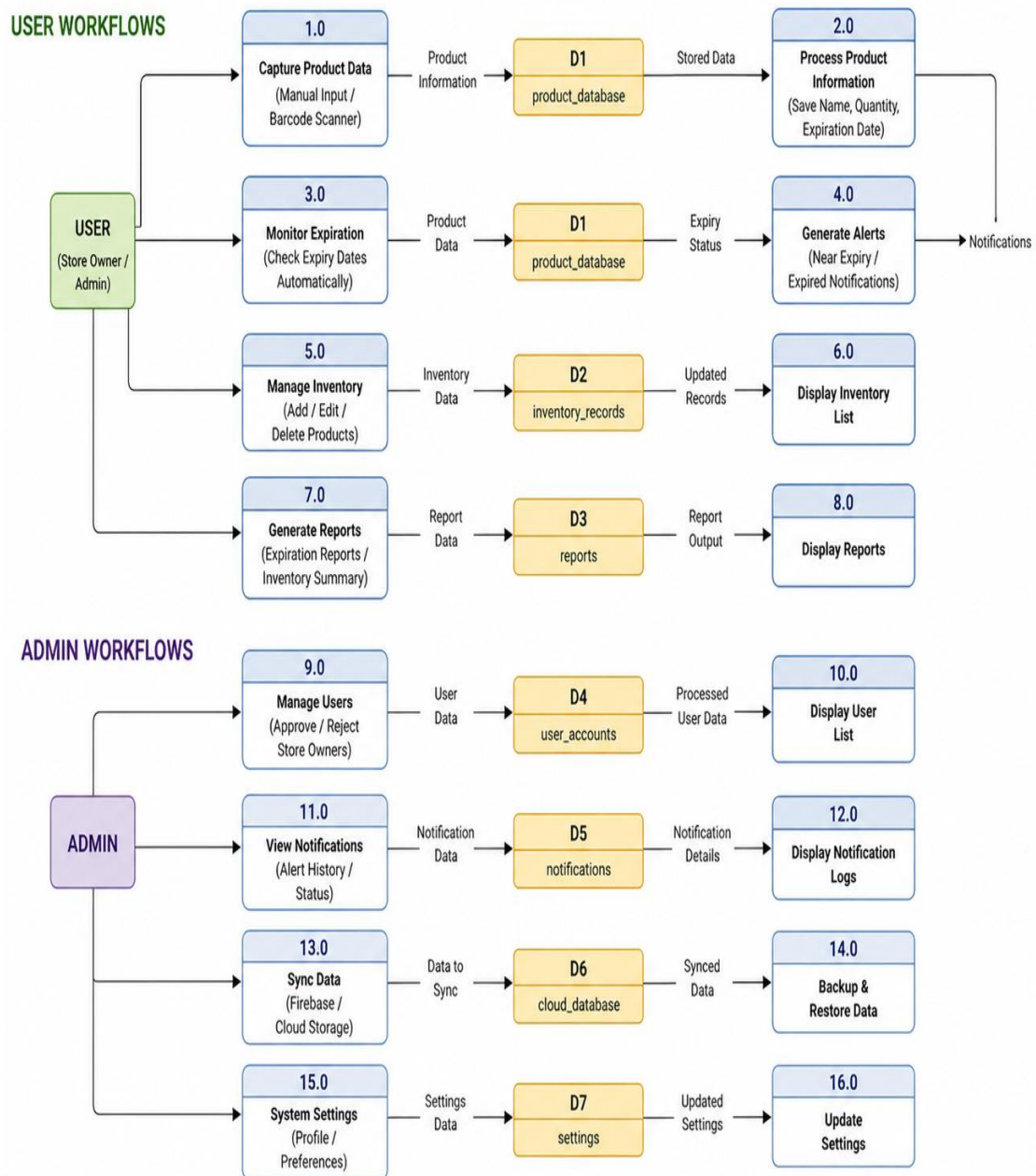


Figure 18 shows the Data Flow Diagram (DFD) of the proposed system, illustrating how data moves within the system and between external entities.

Mockup

A mockup is a visual representation of how a system or application will look and function before it is fully developed. In this Inventory System mockup, it shows how the user interacts with the system, starting from accessing the application, navigating through different modules like Scanner, Inventory, Reports, Profile, and Dashboard, up to receiving alerts and saving data. It helps developers and stakeholders understand the system's layout, flow, and features in a clear and organized way, making it easier to plan, improve, and finalize the design before actual implementation.

Figure 19. User Dashboard

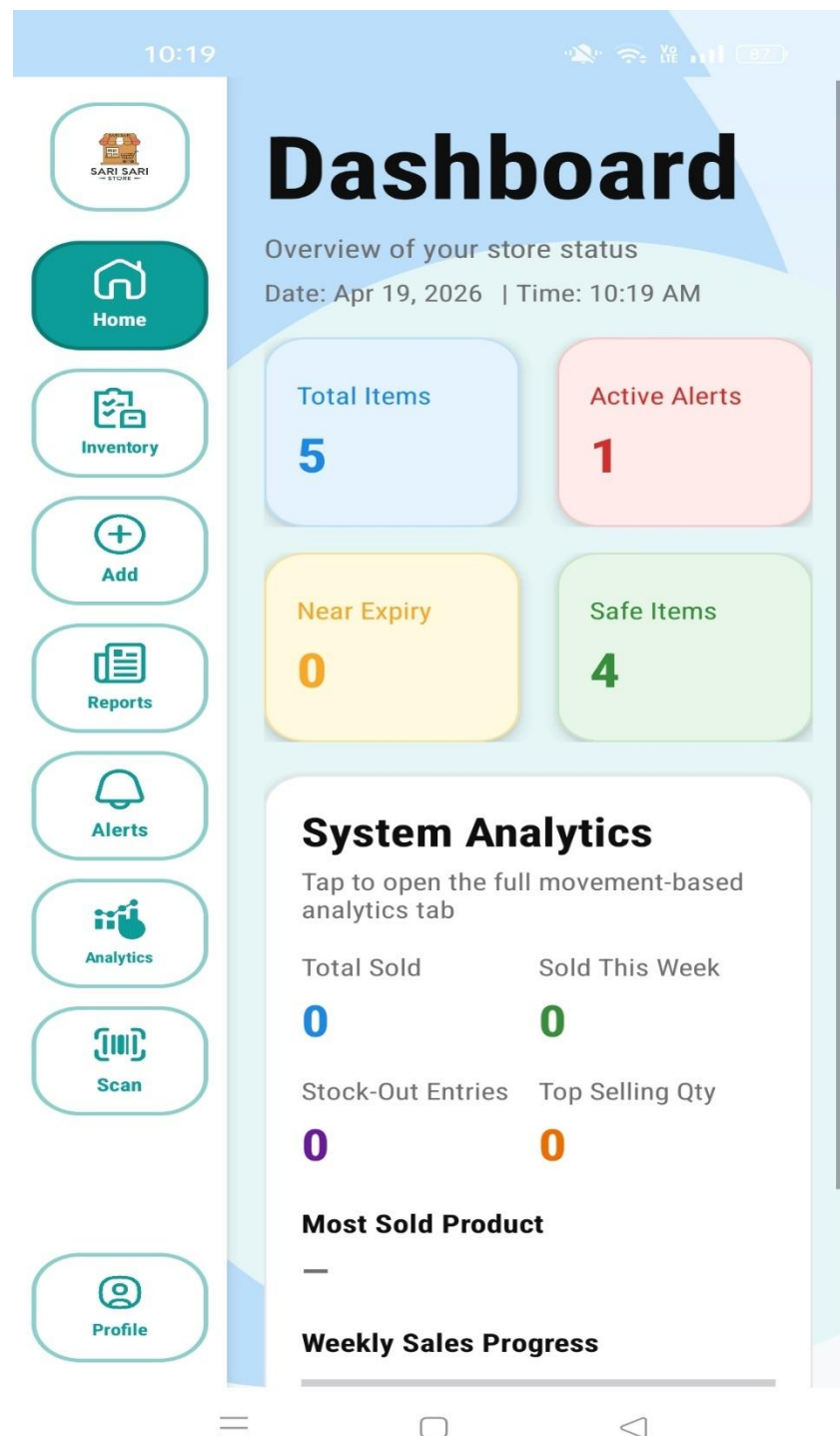


Figure 19 shows the main dashboard of the system, providing an overview of key data and system activities for easy monitoring.

Figure 20. *Barcode Scanner*

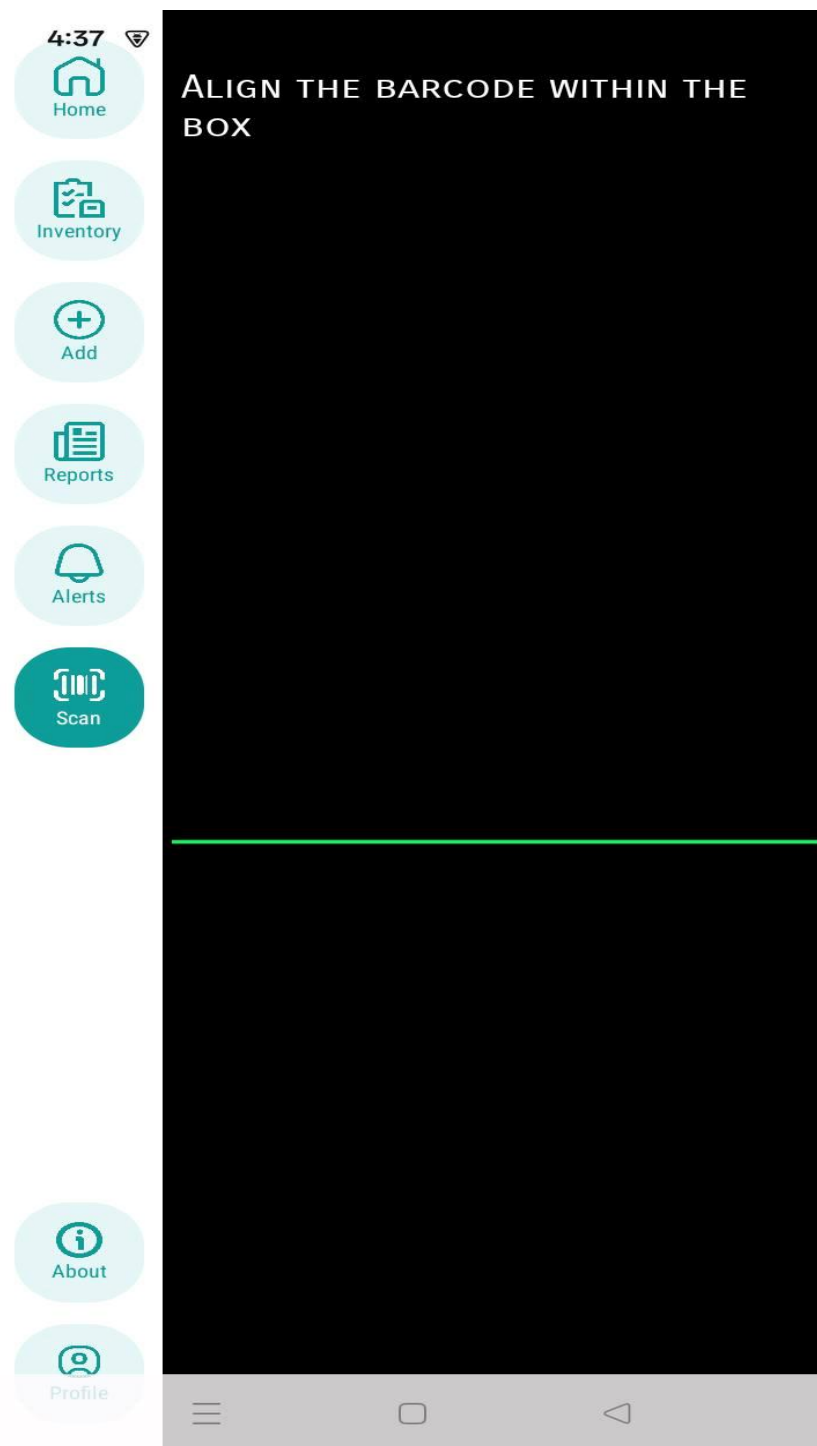


Figure 20 shows the barcode scanner feature used to scan product codes for quick inventory checking and updates.

Figure 21. System Analytics

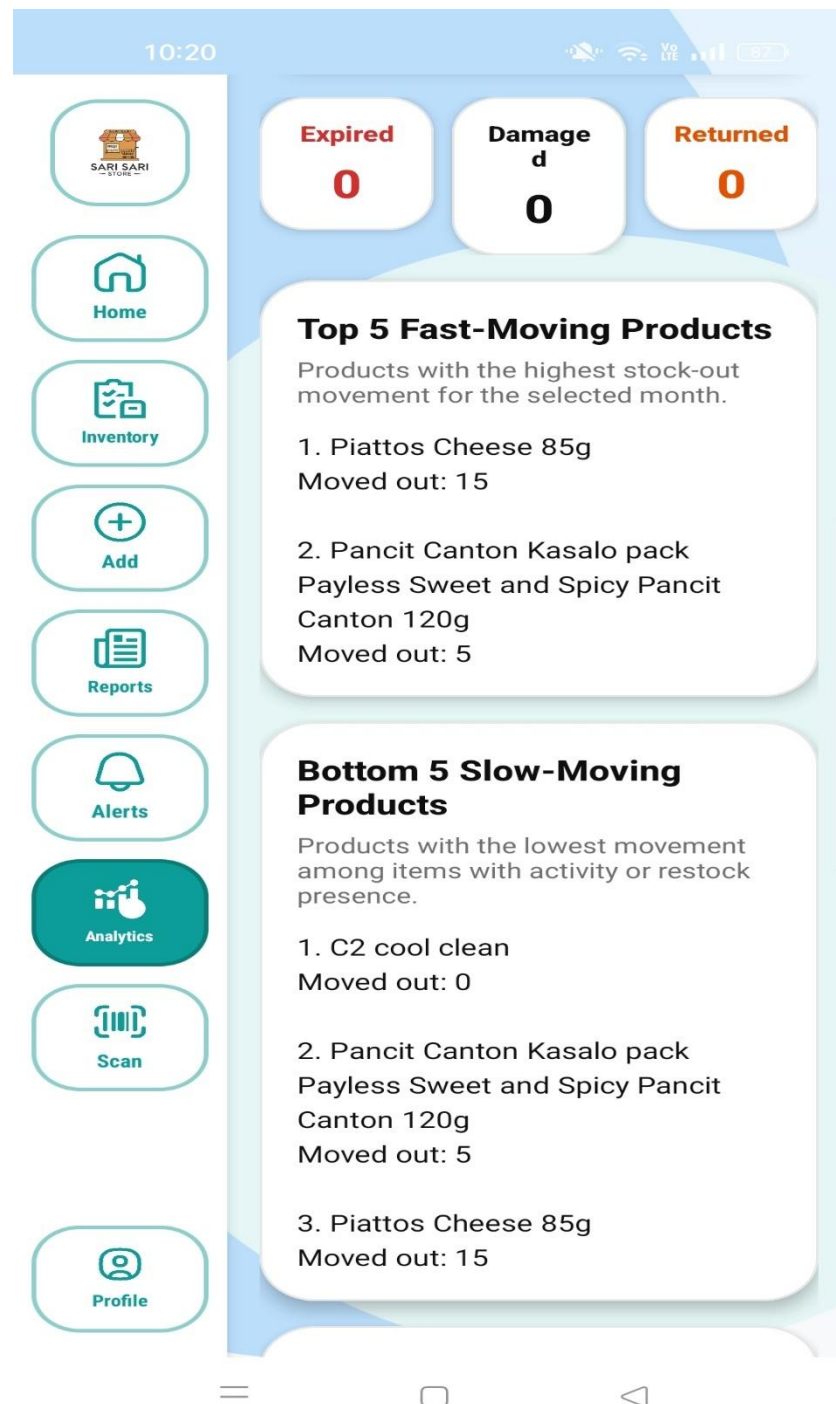
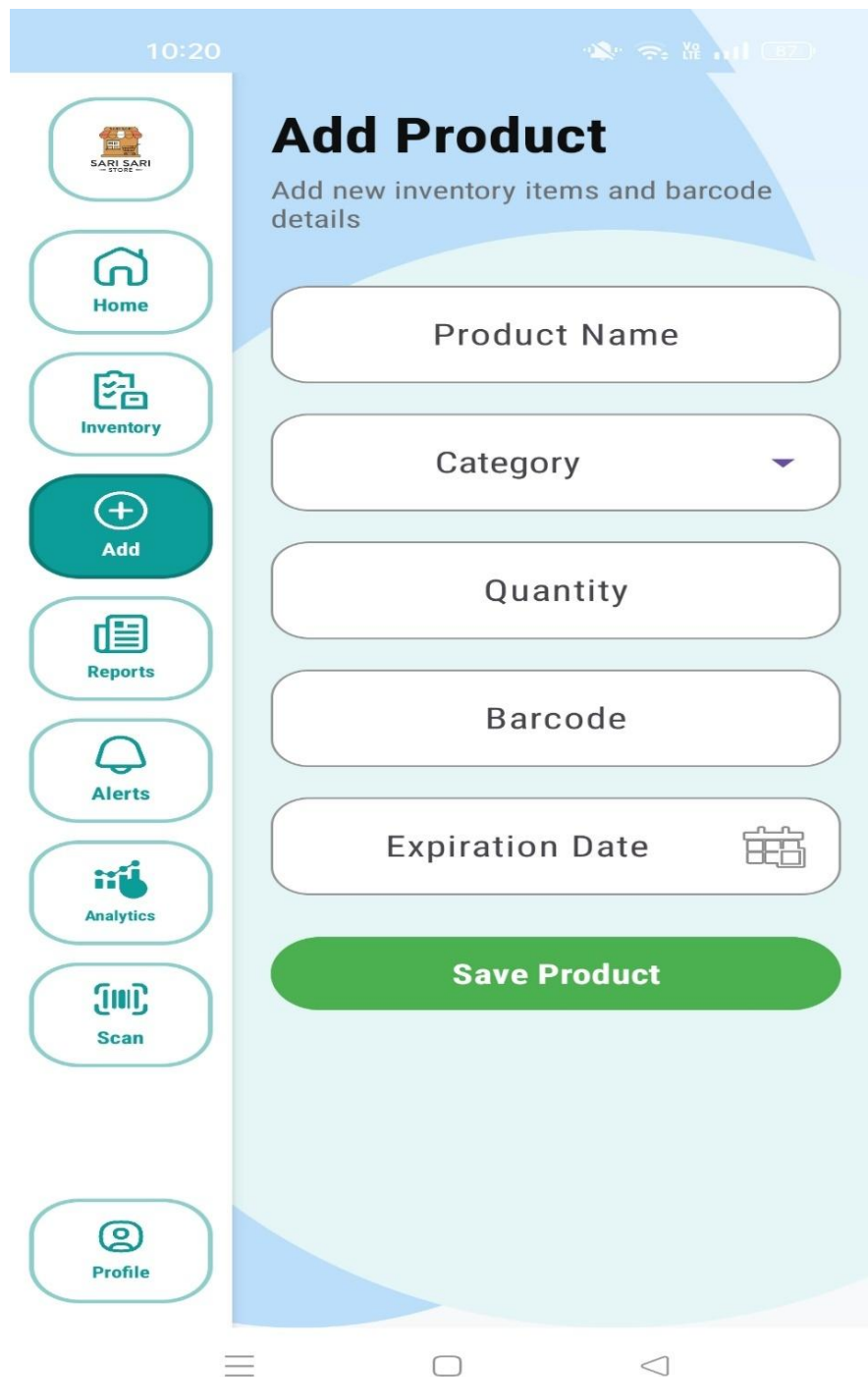


Figure 21 shows the analytics section, displaying graphical data and insights about system performance and store activities

Figure 22. Add Product

10:20

Add Product

Add new inventory items and barcode details

Product Name

Category

Quantity

Barcode

Expiration Date

Save Product

Home

Inventory

Add

Reports

Alerts

Analytics

Scan

Profile

Figure 22 shows the add product interface where users can input and register new product details into the system.

Figure 23. Reports Inventory Logs

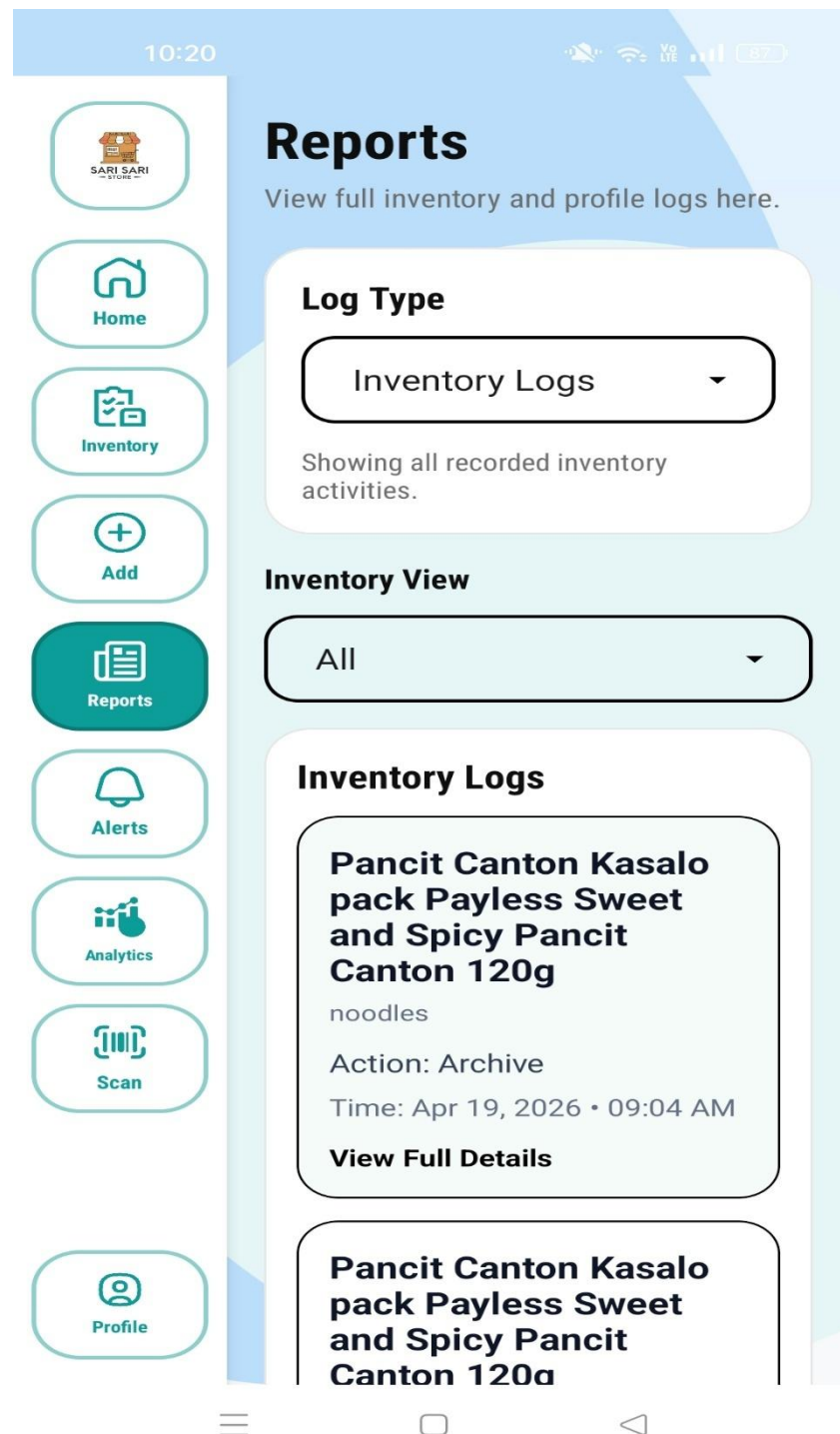


Figure 23 shows the add product interface where users can input and register new product details into the system.

Figure 24. User Profile

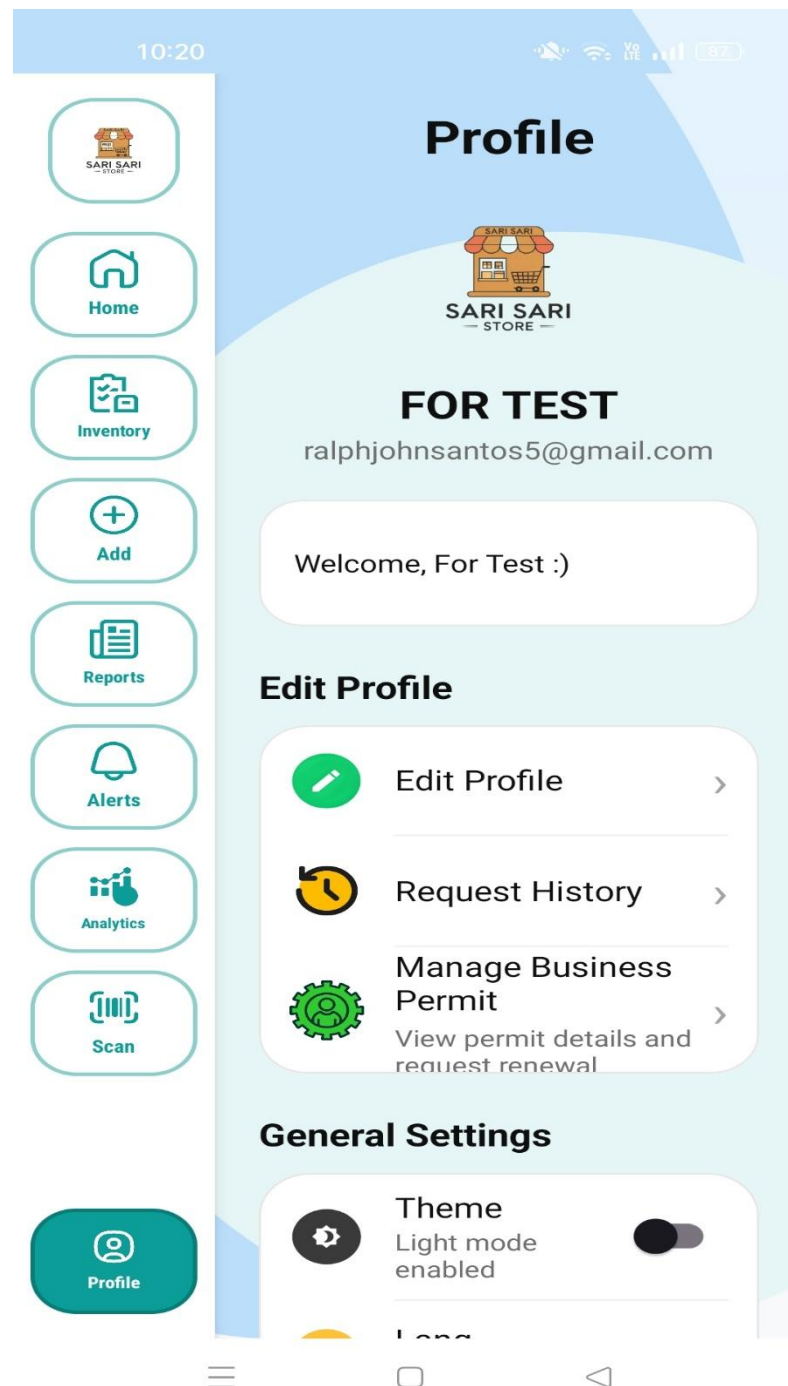
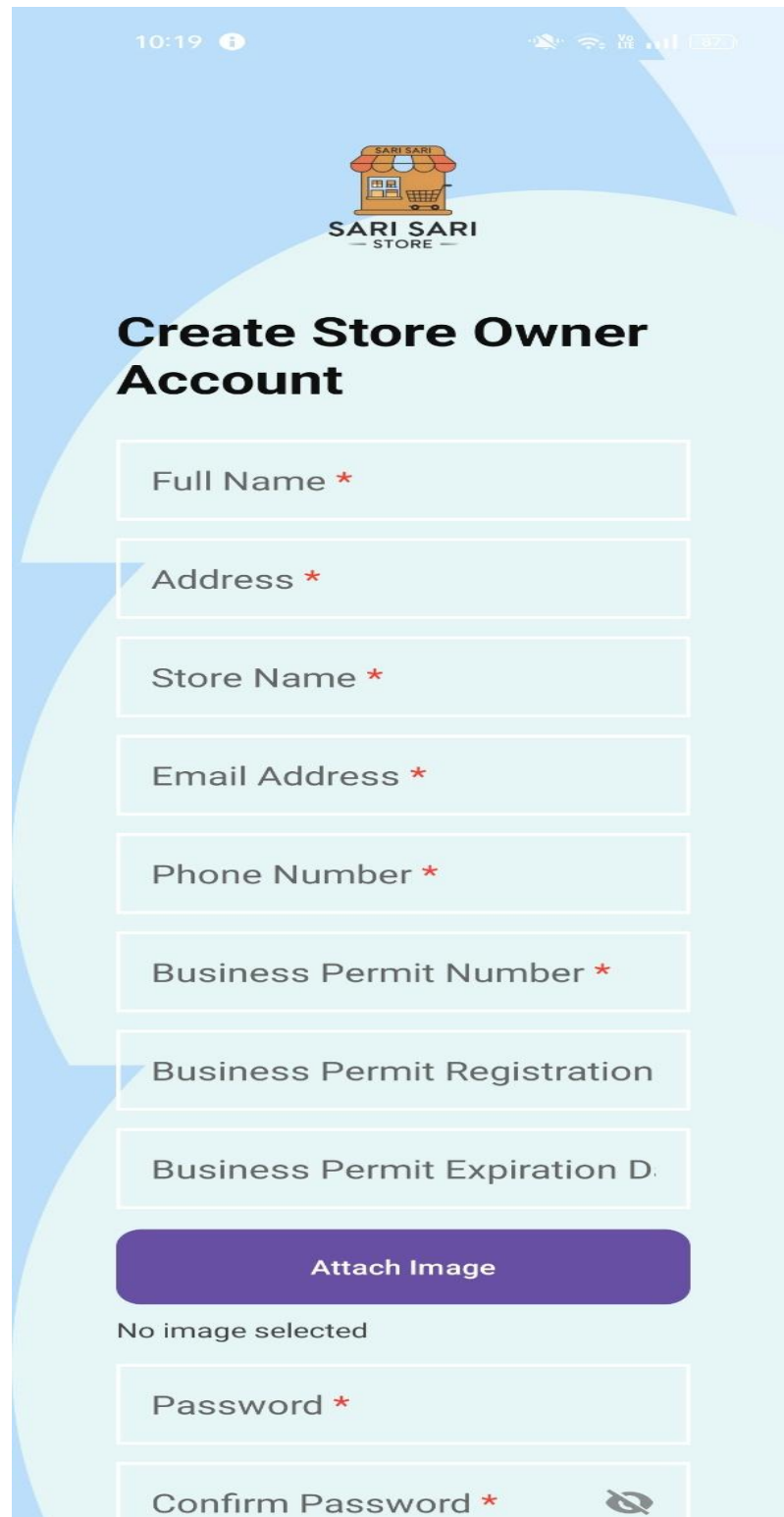


Figure 24 shows the user profile section where account information and personal details are displayed and managed.

Figure 25. *Create Store Owner Account*

10:19 ⓘ

SARI SARI
— STORE —

Create Store Owner Account

Full Name *

Address *

Store Name *

Email Address *

Phone Number *

Business Permit Number *

Business Permit Registration

Business Permit Expiration Date

Attach Image

No image selected

Password *

Confirm Password * 

Figure 25 shows the registration form used to create a new account for system access.

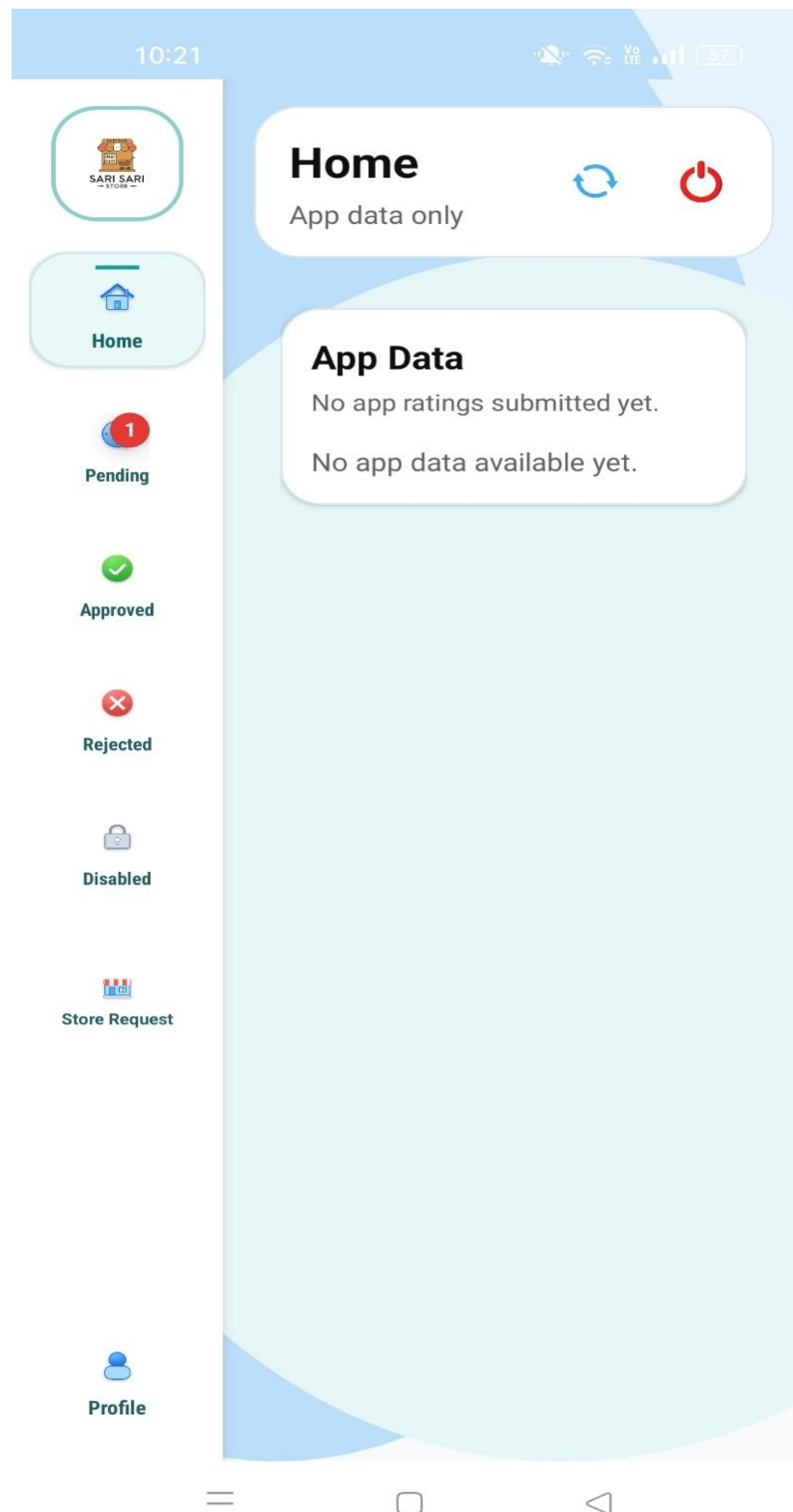
Figure 26. Admin Side Home

Figure 26 shows the admin home page, which provides an overview of system data and quick access to management features.

Figure 27. Admin Side Approved

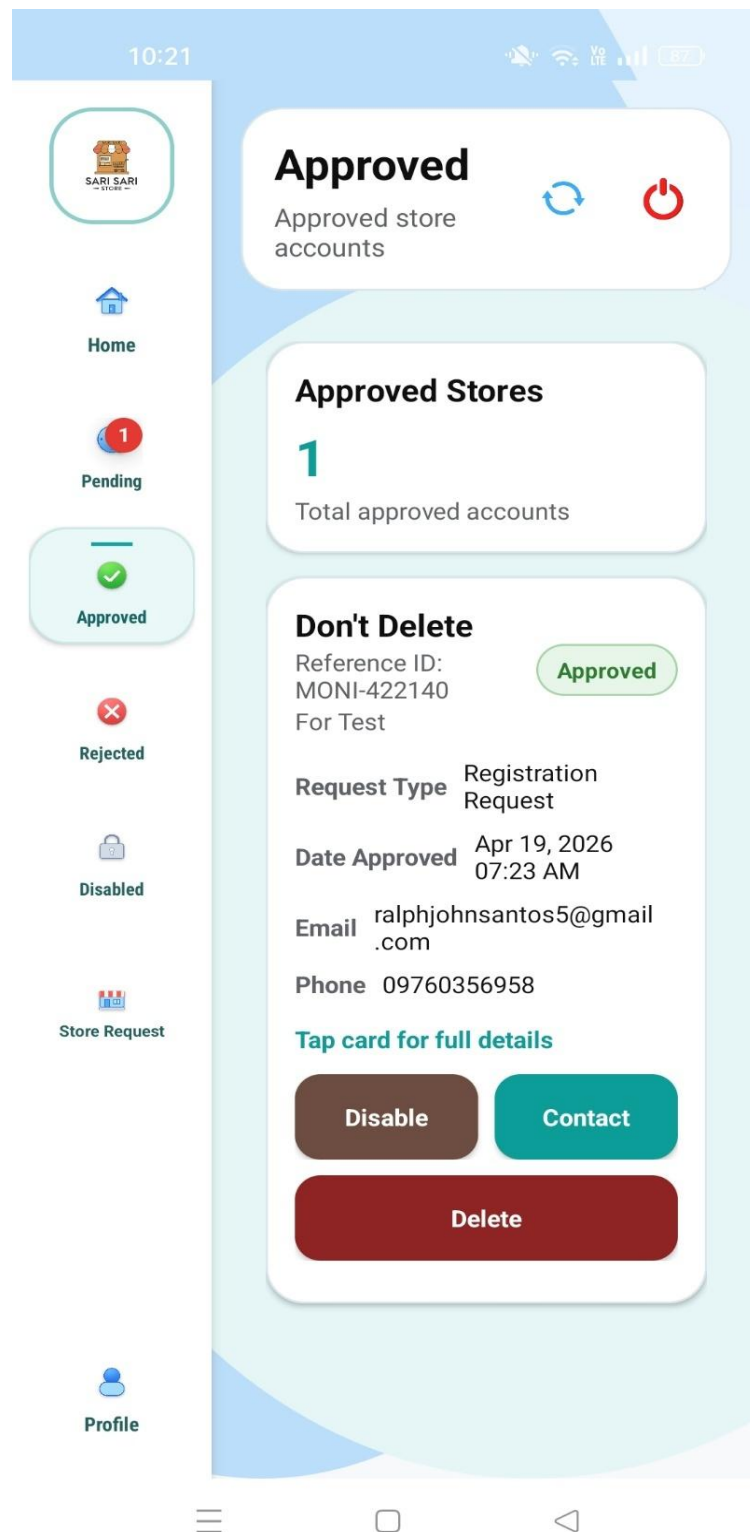


Figure 27 shows the list of approved store owner accounts that are granted system access.

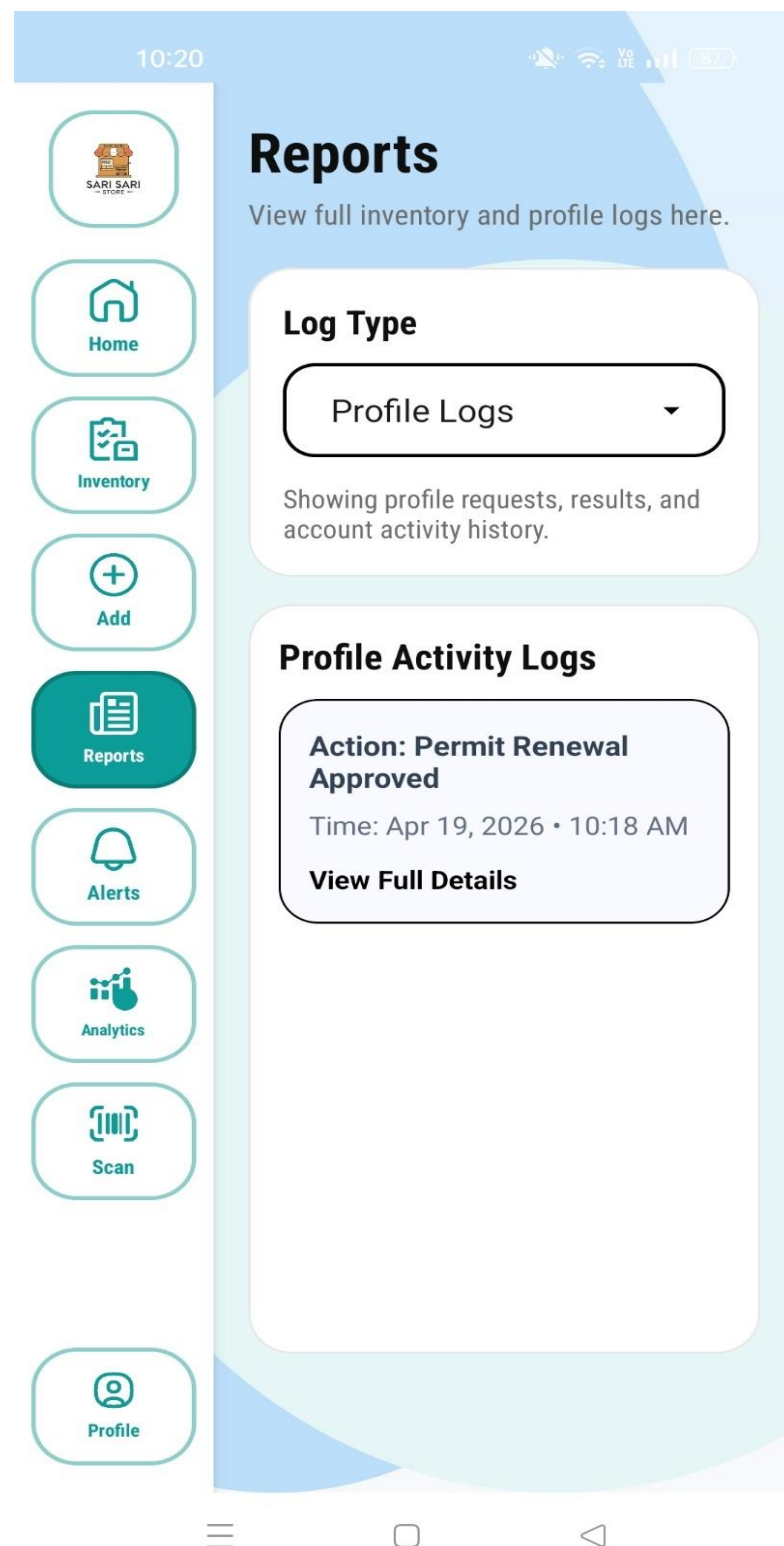
Figure 28. Reports Profile Activity Logs

Figure 28 shows logs of profile-related activities such as updates and changes made by users.

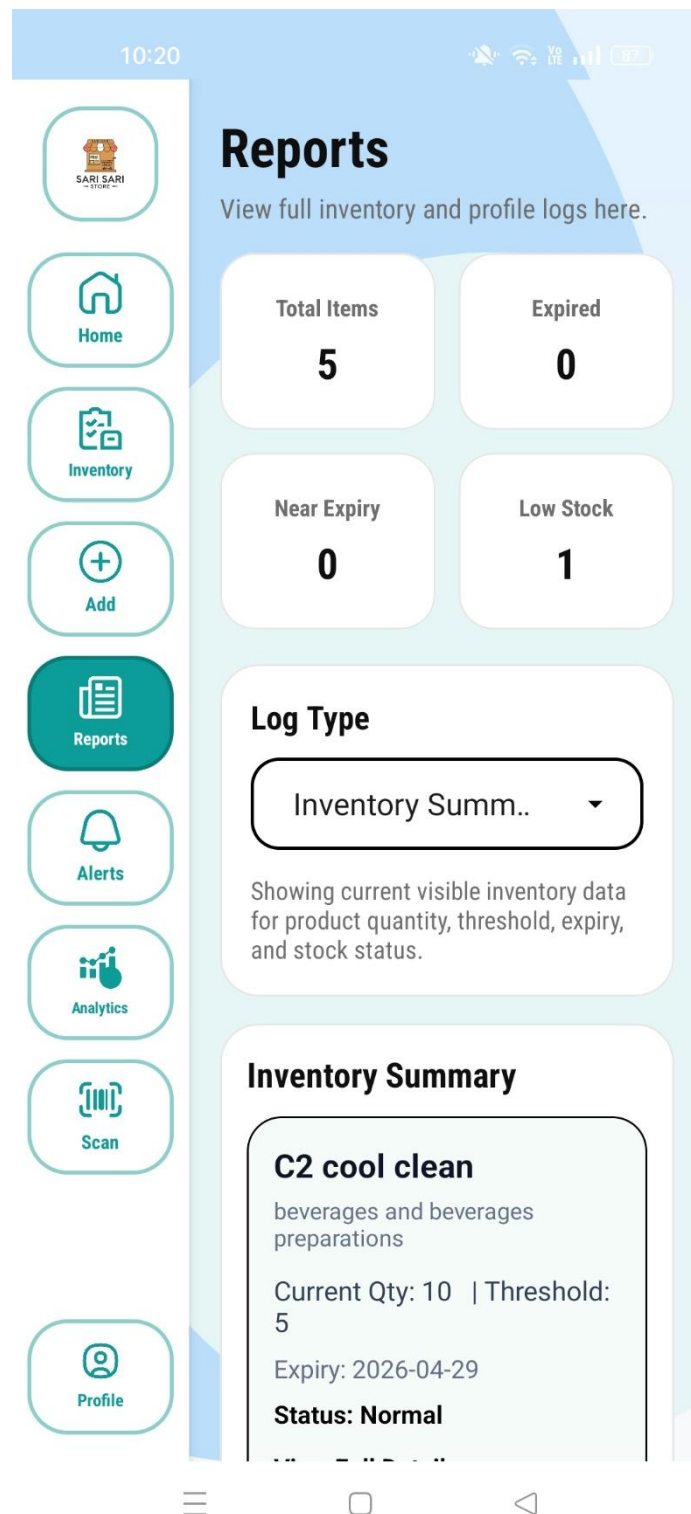
Figure 29. Reports Inventory Summary

Figure 29 shows a summary report of inventory status including stock levels and product distribution.

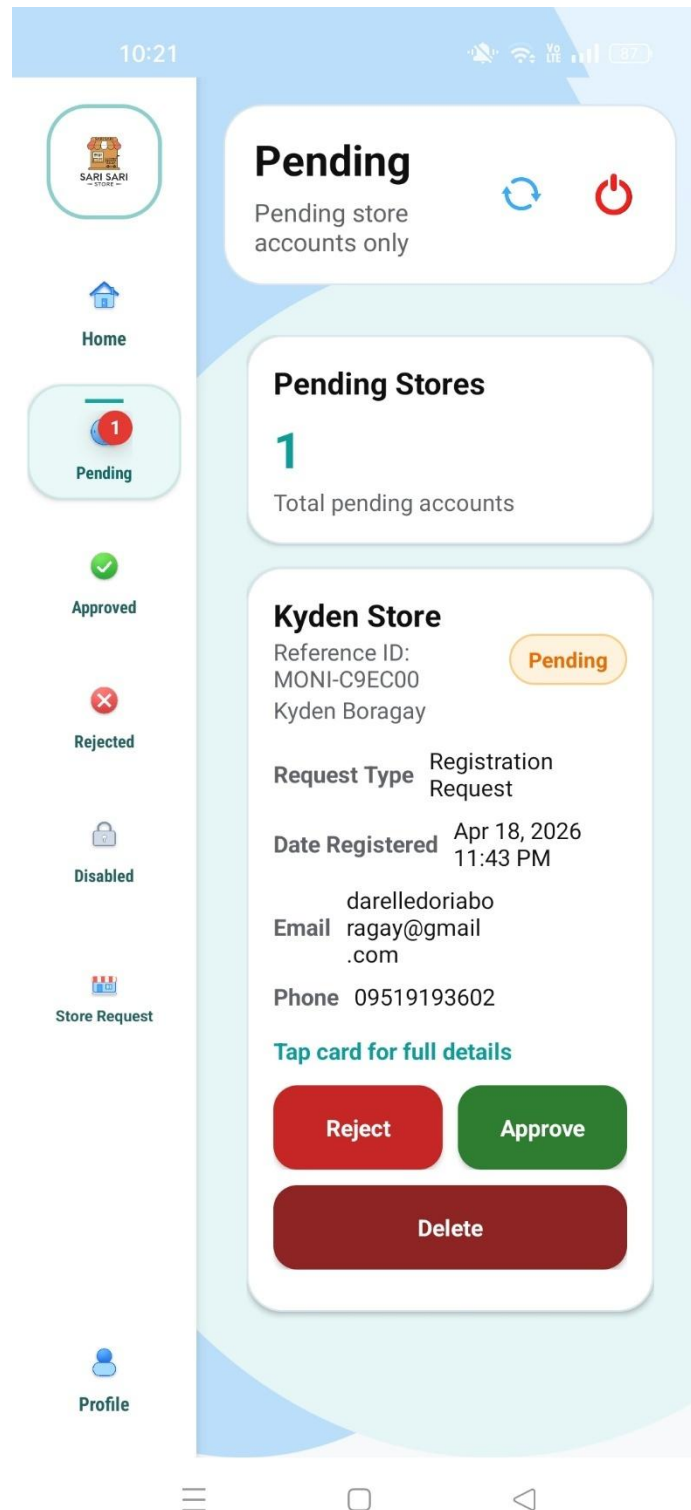
Figure 30. Admin Side Pending Accounts

Figure 30 shows the list of store owner accounts waiting for admin approval.

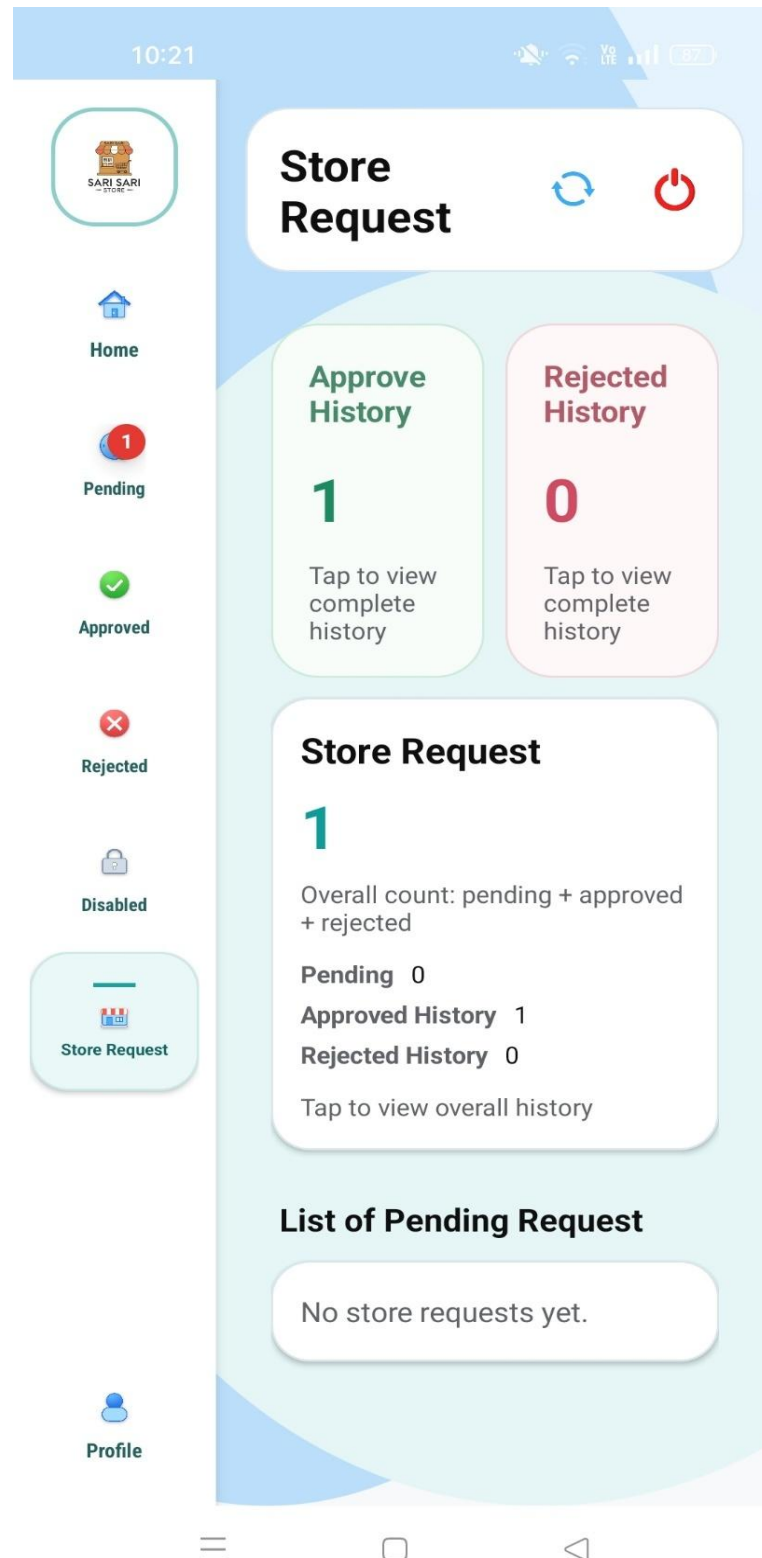
Figure 31. Admin Side Store Owner Request

Figure 31 shows requests submitted by store owners for approval or system changes.

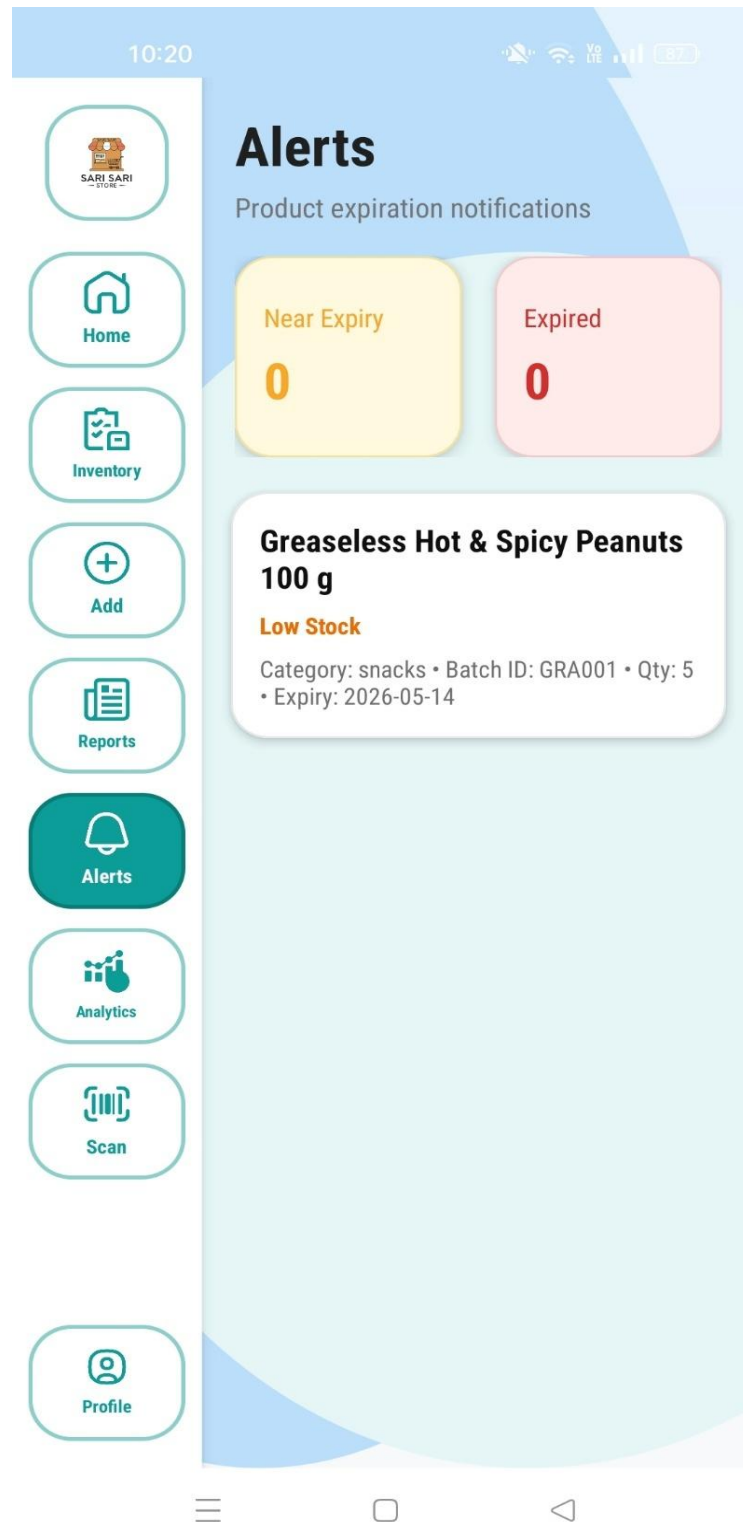
Figure 32. *Product Expiration Alert*

Figure 32 shows the alerts section that notifies users about low stock, expired items, or important system updates.

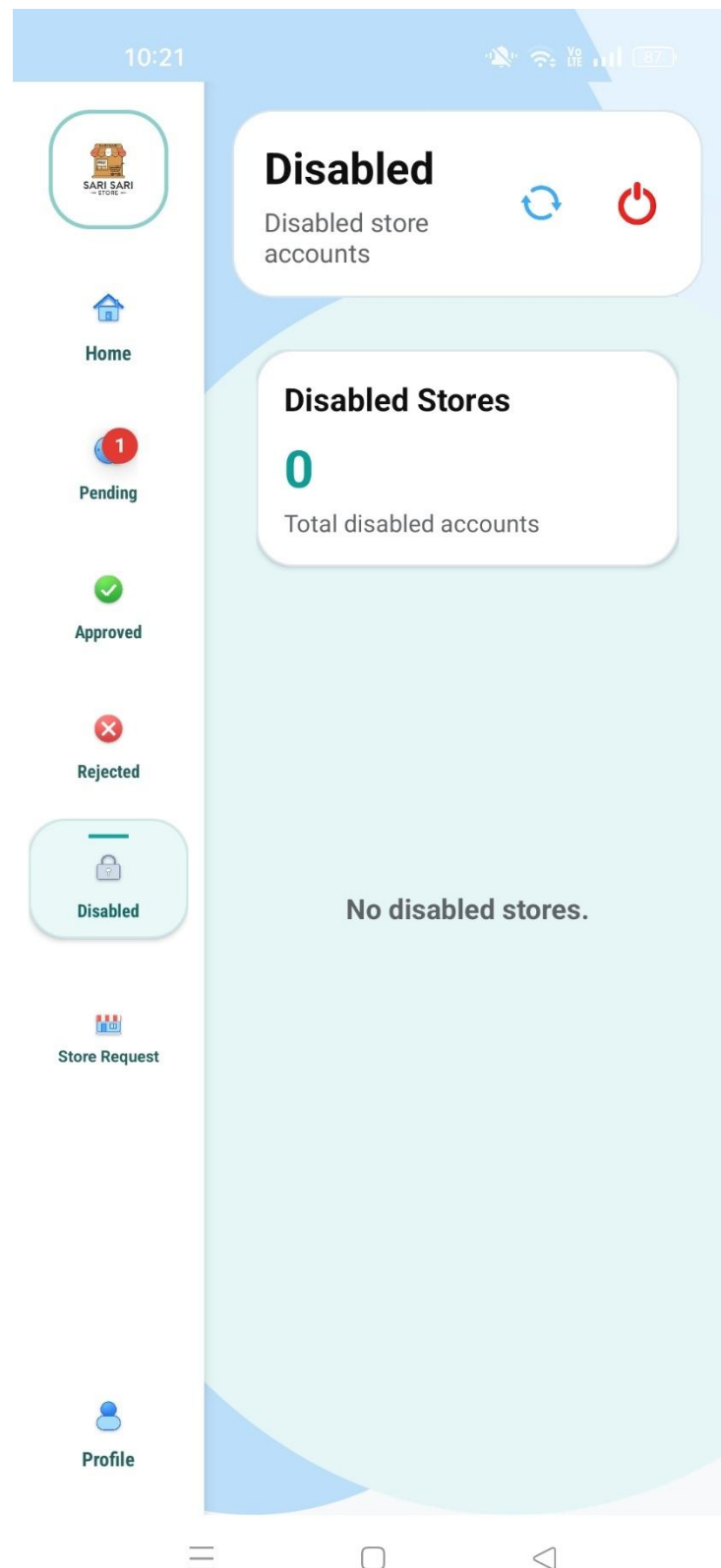
Figure 33. Admin Side Disabled Accounts

Figure 33 shows accounts that have been deactivated or disabled by the admin.

Figure 34. Inventory

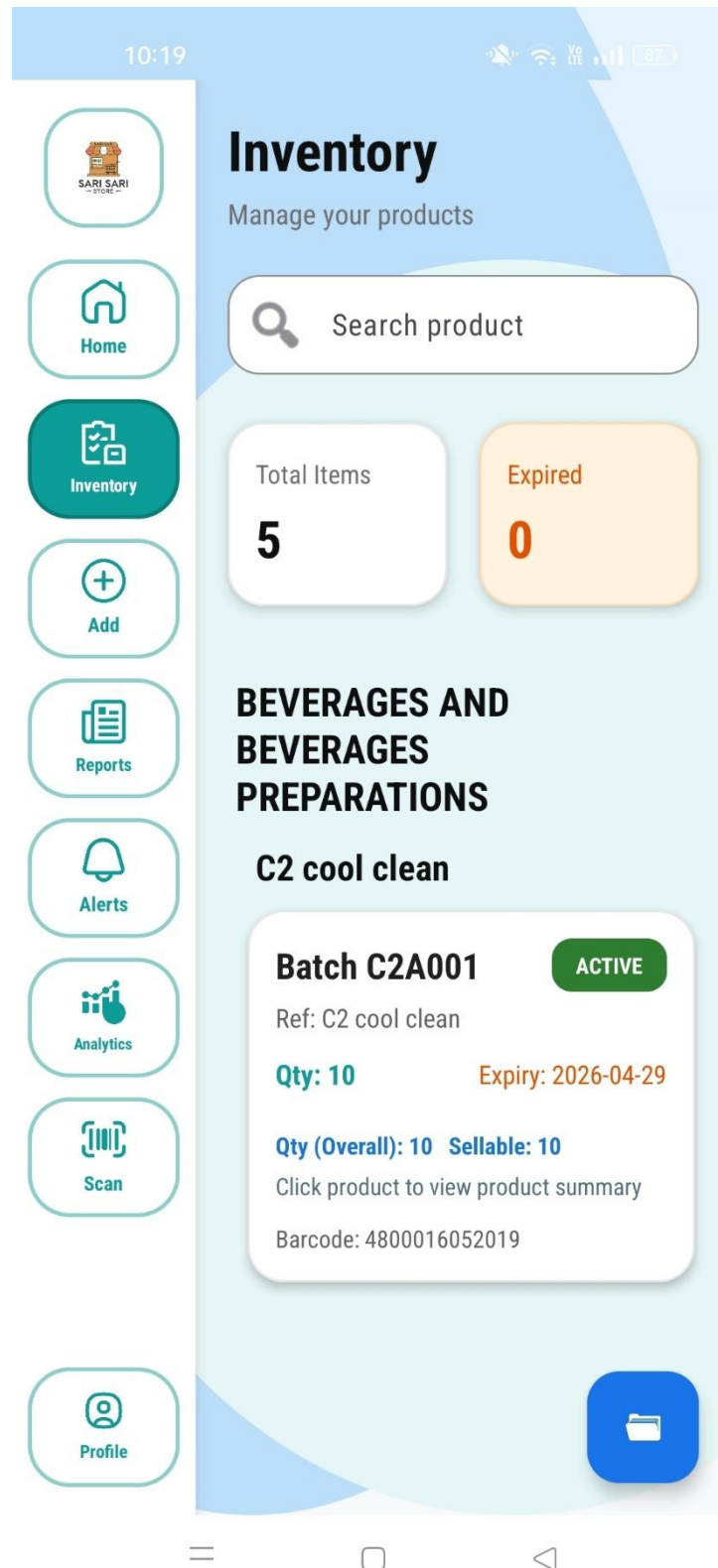
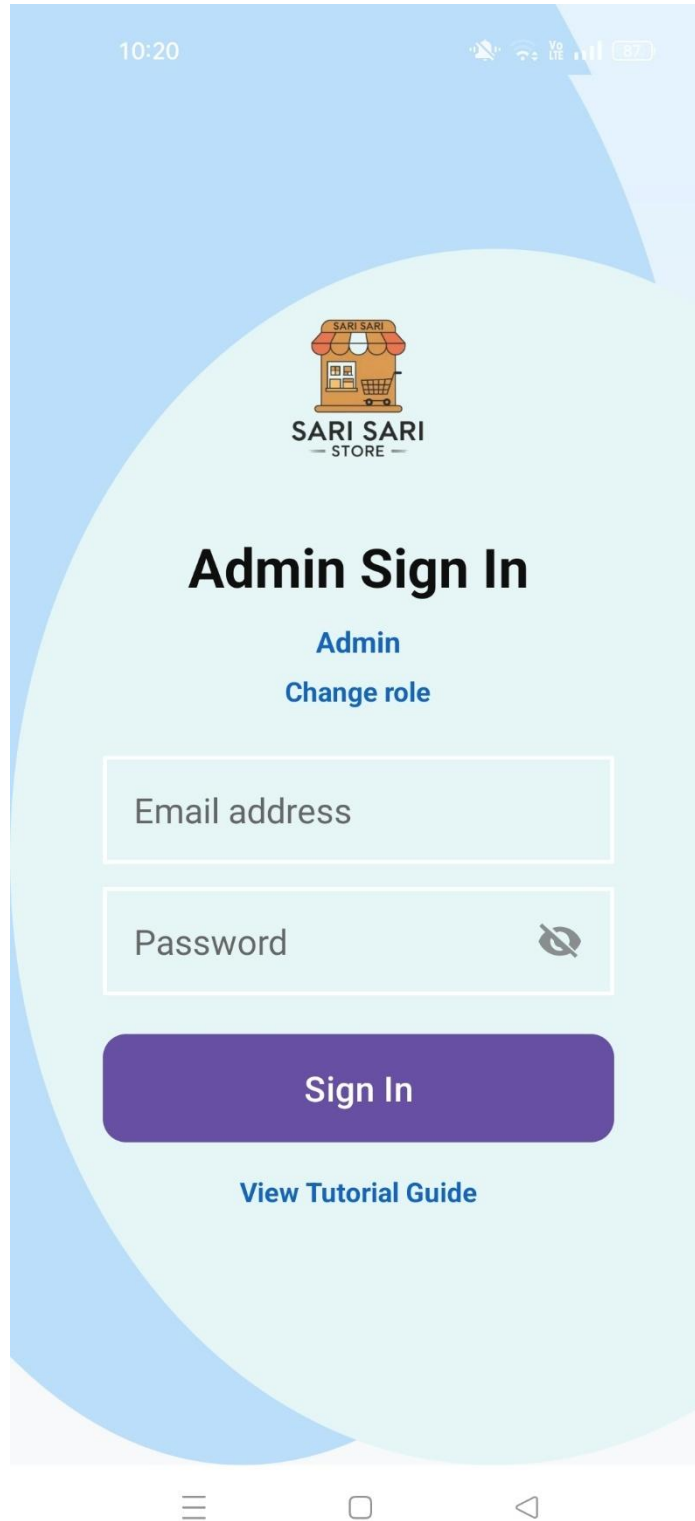


Figure 34 shows the inventory module used to manage and monitor available products and stock levels.

Figure 35. Admin Side Login Form



The image shows a mobile application interface for an admin login. At the top, the status bar displays the time 10:20, signal strength, and battery level at 57%. The app's logo, 'SARI SARI STORE', featuring a small shop icon, is centered. Below the logo, the title 'Admin Sign In' is prominently displayed. Underneath the title, there are two links: 'Admin' and 'Change role'. The login form consists of two input fields: 'Email address' and 'Password'. The 'Password' field has a toggle icon to the right. A large purple 'Sign In' button is positioned below the input fields. At the bottom of the form, there is a link 'View Tutorial Guide'. The entire form is set against a light blue background with a large, stylized circular graphic.

Figure 35 shows the admin login form used for authorized access to the admin dashboard and system controls.

Figure 36. Login Selection Screen

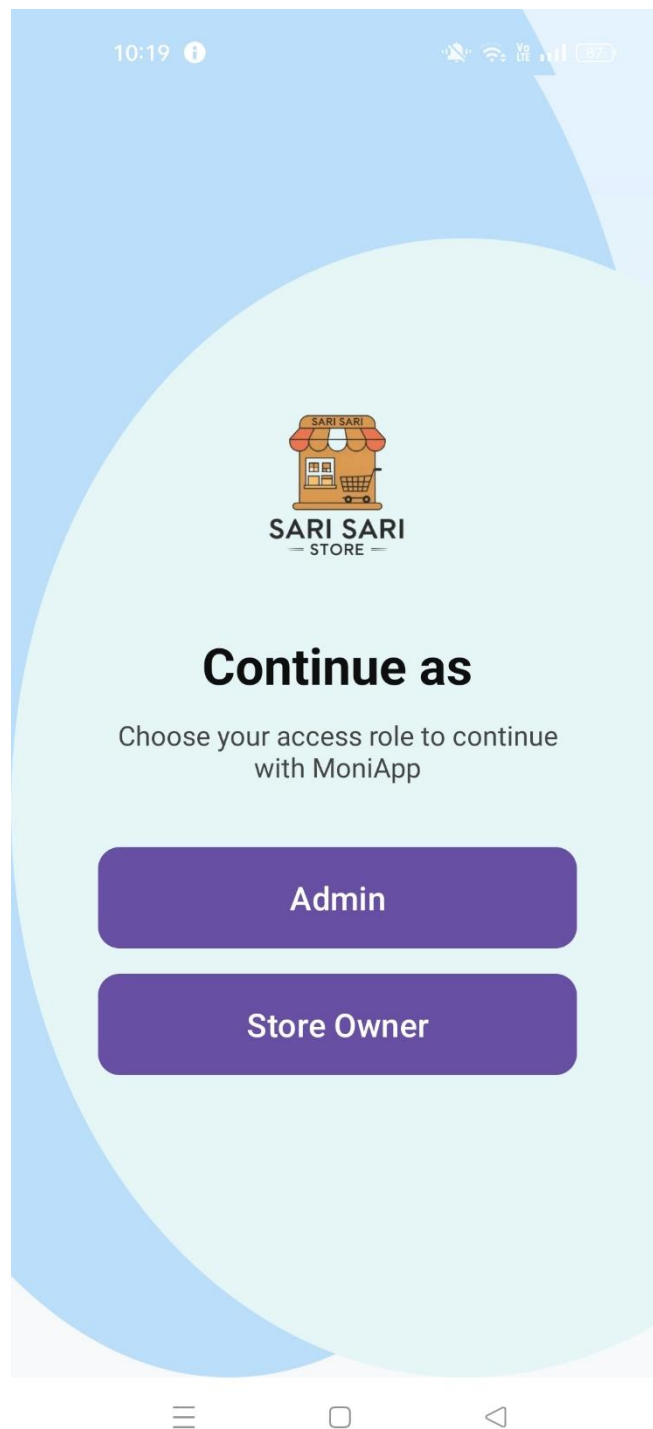


Figure 36 shows the screen where users can choose to continue as admin or store owner before logging in.

Figure 37. User System Analytics

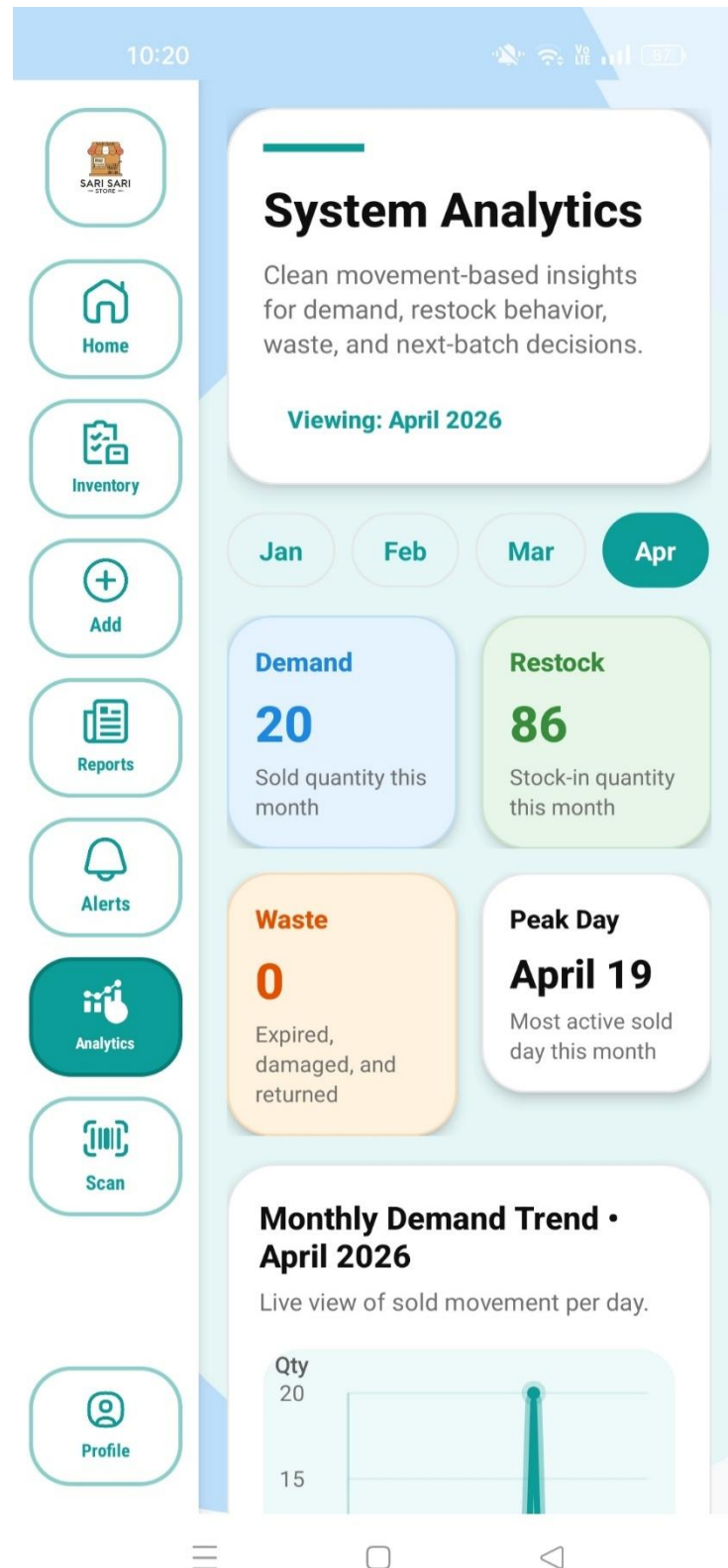


Figure 37 shows the analytics section, displaying graphical data and insights about system performance and store activities.

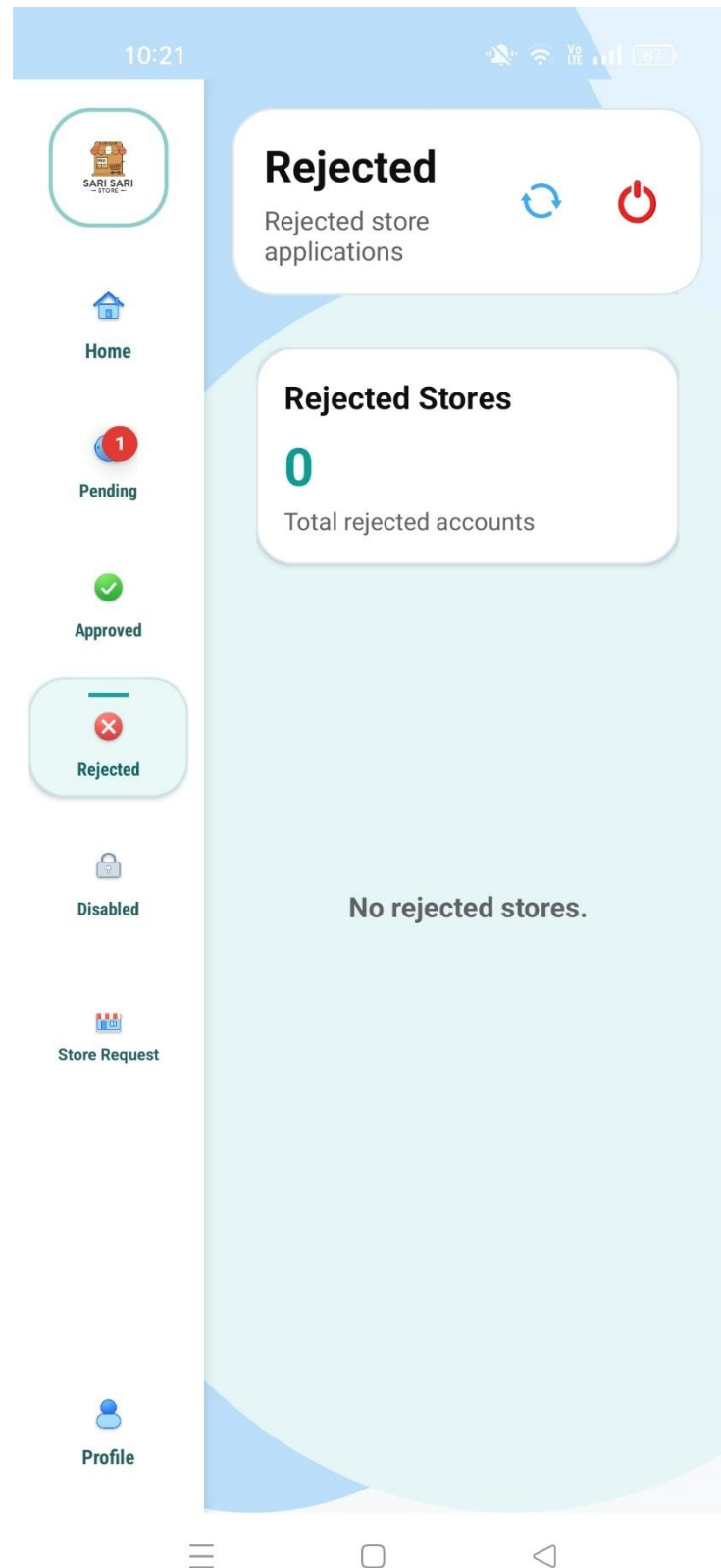
Figure 38. Admin Side Rejected

Figure 38 shows store owner accounts that have been rejected by the admin.

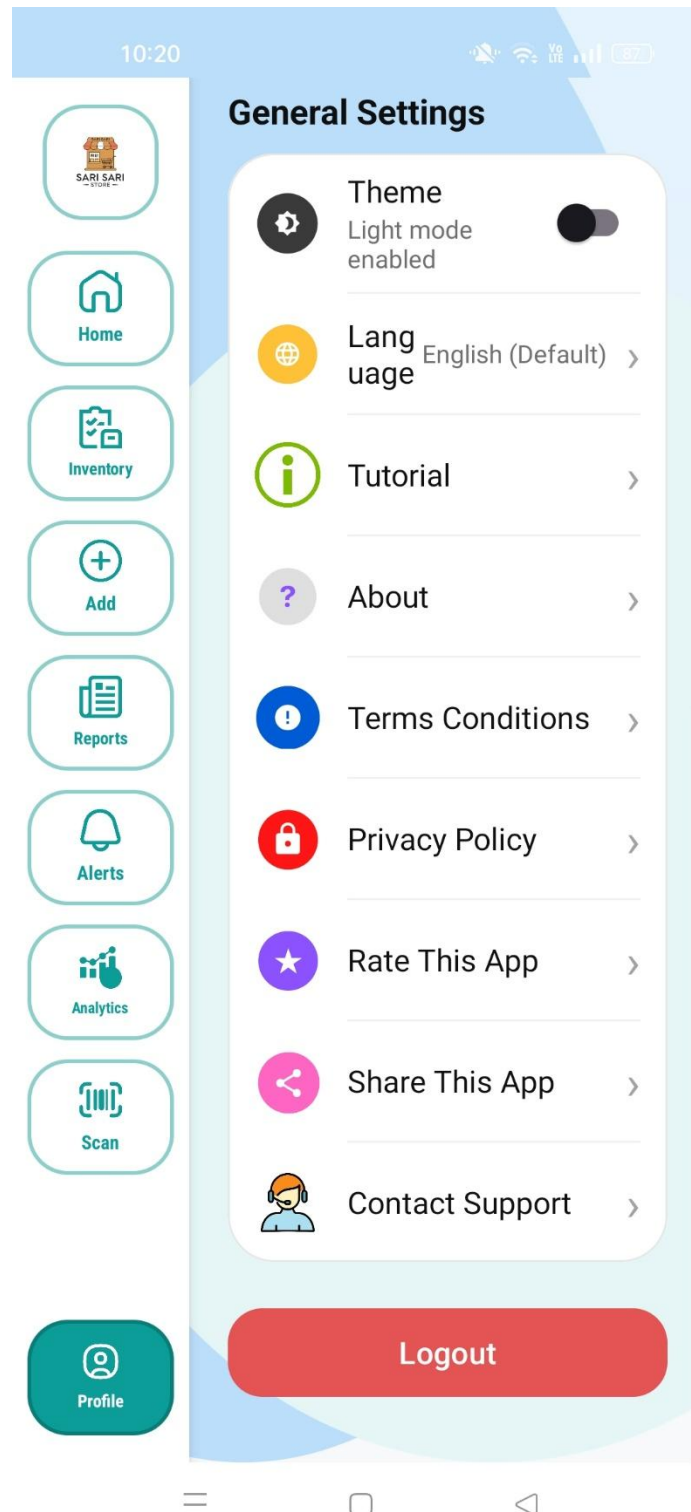
Figure 39. *General Settings of Store Owner*

Figure 39 shows the system settings section used to configure general application preferences.

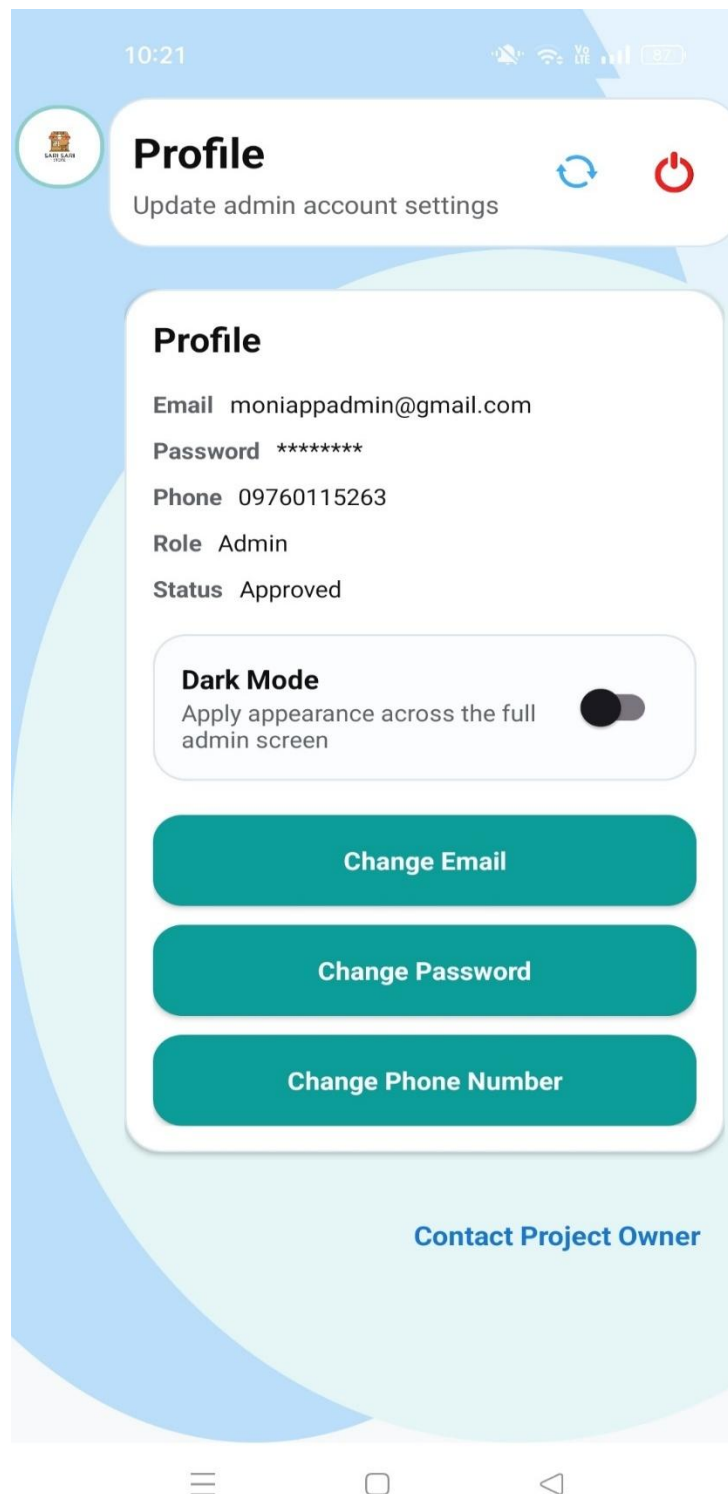
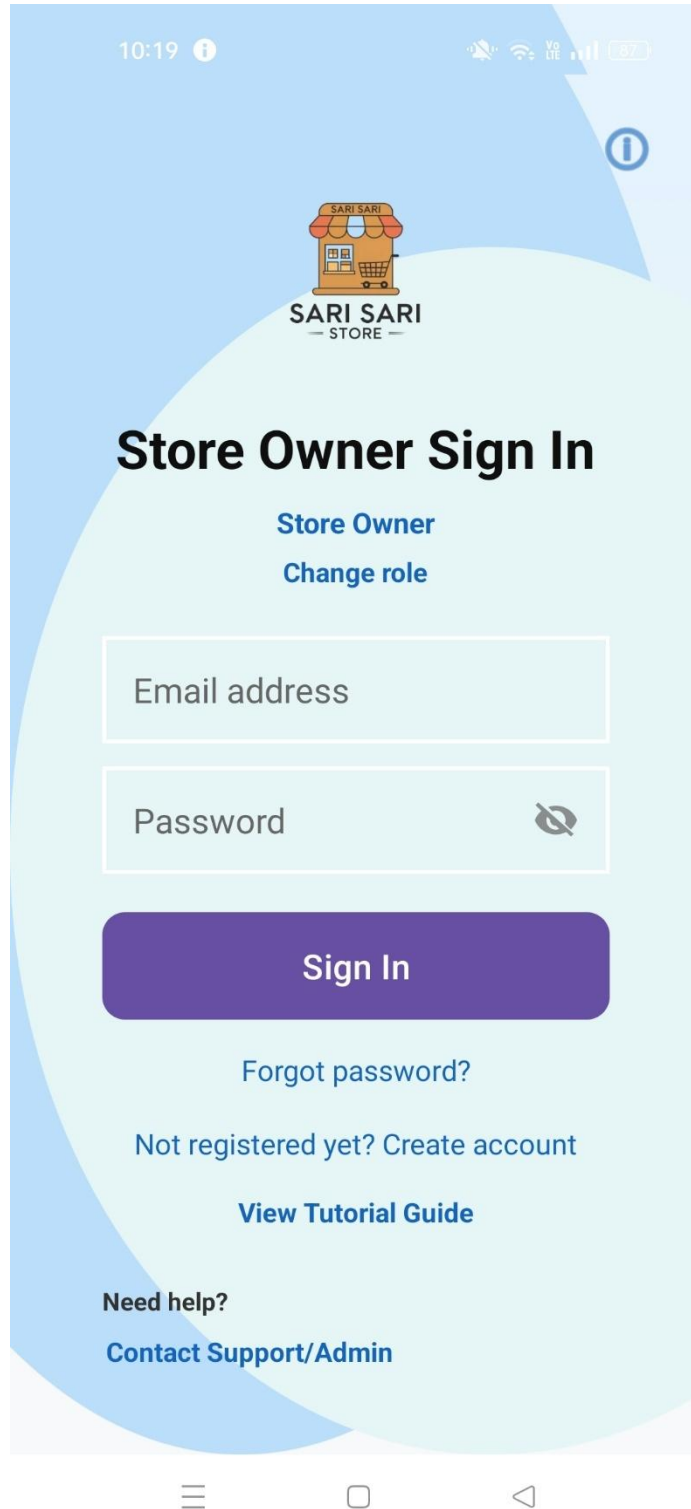
Figure 40. Admin Profile Account Settings

Figure 40 shows the admin profile settings where account details can be updated and managed.

Figure 41. *Store Owner Login Form*

10:19 ⓘ

🔔 📶 4G LTE 📶 🔋 87%

ⓘ


SARI SARI
— STORE —

Store Owner Sign In

[Store Owner](#)
[Change role](#)

Email address

Password 🔒

Sign In

[Forgot password?](#)

[Not registered yet? Create account](#)

[View Tutorial Guide](#)

Need help?
[Contact Support/Admin](#)

Figure 41 shows the login form for store owners to securely access their account in the system.

Figure 42. IT Experts evaluating the system



Figure 43. IT Experts evaluating the system



Figure 42 and 43 shows IT experts evaluating the system to ensure its performance, reliability, and security meet required standards.

Figure 44. Residence Evaluating System



Figure 45. Residence Evaluating System

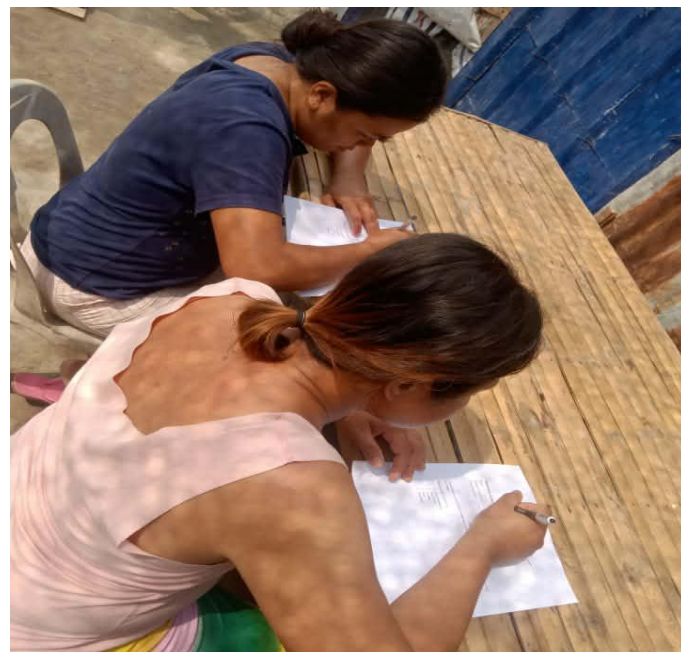


Figure 44 and 45 shows Residence of Barangay Palihan evaluating the system to ensure its performance, reliability, and security meet required standards.

Figure 46. Certificate of Deployment



Figure 46 presents the certificate of deployment, confirming that the system has been successfully implemented and is operational in its intended environment.

Develop

This section answers the development-related questions from the Statement of the Problem.

1. What are the technologies used in developing the Mobile-Based Product Expiration Monitoring Application?

The system was developed using the following tools and technologies:

Android Studio – Used as the primary Integrated Development Environment (IDE) for building the Android mobile application.

Java and Kotlin – Programming languages used for developing the core functionality of the application.

XML – Used for designing the graphical user interface and application layouts.

Firebase Firestore – A cloud-based database used to store product information and expiration records in real time.

Cloud Storage – Used to securely store data and allow access across multiple devices.

Barcode Technology – Used to quickly input product information by scanning product codes.

Review

The review process of the Mobile-Based Product Expiration Monitoring Application for Sari-Sari Store in Palihan, Hermosa, Bataan is important in checking if the system works properly and meets the needs of the store owners. The evaluation focused on different aspects such as functionality, usability, reliability, performance, and overall system quality using the ISO/IEC 25010 standard. It also looked at the user interface, accuracy of tracking expiration dates, notification alerts, barcode scanning, and data storage using Firebase Firestore. Based on the results, the system helped improve the way store owners monitor their products by giving timely alerts and reducing errors from manual recording. It also made inventory management easier and more organized. Some suggestions were also gathered to further improve the system. Overall, the review showed that the application is useful, reliable, and effective in helping sari-sari store owners manage their products and avoid selling expired items.

Profile of the Respondents

Table 1. *Profile of the Respondents*

PROFESSION	FREQUENCY	PERCENTAGE
RESIDENTS	99	90.00%
IT Experts	10	9.09%
Administrator	1	0.91%
TOTAL	110	100%

Table 1 shows that out of 110 respondents, the majority were residents with 90.00%, followed by IT experts at 9.09%, while the administrator comprised 0.91%. This distribution of respondents is appropriate for the study as it represents both the primary users of the system and the technical evaluators involved in assessing its overall performance.

Functionality Evaluation

The evaluation of the system was conducted based on the ISO/IEC 25010 software quality standards, which provide a framework for assessing the quality of software products. The system was evaluated using key quality characteristics including Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

These criteria were used to determine how well the system performs in terms of meeting user requirements, ensuring efficient operation, and maintaining overall system quality.

Statistical Tools Used

To analyze the results of the study, the following statistical methods were applied:

Frequency – This method was used to determine the number of respondents who participated in the evaluation of the system.

Percentage – This was used to compute the proportion of respondents under specific categories. The formula used is: $P = (F \times 100) / N$, where: P = Percentage, F = Frequency, N = Total number of respondents

Weighted Mean – This statistical tool was used to determine the average rating of the respondents in evaluating the system based on the given criteria, providing an overall assessment of the system's performance.

Table 2. Likert Scale

RANK	NUMERICAL SCALE	INTERPRETATION
5	4.51 - 5.00	Very Satisfactory
4	3.51 - 4.50	Satisfactory
3	2.51 - 3.50	Average
2	1.51 - 2.50	Poor
1	1.00 - 1.50	Very Poor

Likert Scale – The findings were interpreted using the Likert Scale, a widely recognized rating system typically employed in surveys and questionnaires. This scale is designed to gauge respondents' attitudes, emotions, and degrees of agreement or disagreement with specific statements, offering valuable insights across a range of assessment scenarios.

Table 3. Overall System Evaluation

	Rating N = 110											
CRITERIA	5	%	4	%	3	%	2	%	1	%	MEAN	
	FUNCTIONAL SUITABILITY											
Functional Completeness	92	83.64	14	12.73	3	2.73	1	0.91	0	0	4.79	
Functional Correctness	95	86.36	12	10.91	2	1.82	1	0.91	0	0	4.82	
Functional Appropriateness	94	85.45	13	11.82	2	1.82	1	0.91	0	0	4.81	
	General weighted mean based on functional suitability											4.81
	PERFORMANCE EFFICIENCY											
Time Behavior	88	80	17	15.45	3	2.73	2	1.82	0	0	4.74	
Resource Utilization	86	78.18	18	16.36	4	3.64	2	1.82	0	0	4.71	
Capacity	87	79.09	16	14.55	4	3.64	3	2.73	0	0	4.7	
	General weighted mean based on Performance Efficiency											4.72
	COMPATIBILITY											
Co-existence	78	70.91	22	20	6	5.45	4	3.64	0	0	4.58	
Interoperability	82	74.55	20	18.18	5	4.55	3	2.73	0	0	4.64	
	General weighted mean based on Compatibility											4.61
	USABILITY											
Recognizability	89	80.91	15	13.64	4	3.64	2	1.82	0	0	4.74	
Learnability	90	81.82	14	12.73	4	3.64	2	1.82	0	0	4.75	
Operability	91	82.73	13	11.82	4	3.64	2	1.82	0	0	4.76	
User Error Protection	75	68.18	23	20.91	7	6.36	5	4.55	0	0	4.53	
UI Aesthetics	88	80	16	14.55	4	3.64	2	1.82	0	0	4.74	
Accessibility	89	80.91	15	13.64	4	3.64	2	1.82	0	0	4.74	
	General weighted mean based on Usability											4.71
	RELIABILITY											
Maturity	92	83.64	14	12.73	3	2.73	1	0.91	0	0	4.79	
Availability	94	85.45	13	11.82	2	1.82	1	0.91	0	0	4.81	
Fault Tolerance	80	72.73	22	20	5	4.55	3	2.73	0	0	4.63	
	General weighted mean based on Reliability											4.71
	SECURITY											
Confidentiality	85	77.27	19	17.27	4	3.64	2	1.82	0	0	4.7	
Integrity	83	75.45	20	18.18	5	4.55	2	1.82	0	0	4.67	
	General weighted mean based on Security											4.69
	MAINTAINABILITY											
Modularity	88	80	16	14.55	4	3.64	2	1.82	0	0	4.74	
Reusability	80	72.73	20	18.18	6	5.45	4	3.64	0	0	4.61	
Analyzability	82	74.55	18	16.36	6	5.45	4	3.64	0	0	4.6	
Modifiability	86	78.18	17	15.45	4	3.64	3	2.73	0	0	4.7	
Testability	87	79.09	16	14.55	4	3.64	3	2.73	0	0	4.7	
	General weighted mean based on Maintainability											4.67
	PORABILITY											
Adaptability	85	77.27	18	16.36	4	3.64	3	2.73	0	0	4.69	
Installability	88	80	16	14.55	4	3.64	2	1.82	0	0	4.74	
Replaceability	90	81.82	14	12.73	4	3.64	2	1.82	0	0	4.75	
	General weighted mean based on Portability											4.73
	GENERAL WEIGHTED MEAN											4.71

Based on the evaluation criteria presented in Table 3, the majority of respondents rated the system as Very Satisfactory with an overall General Weighted Mean of 4.71. These criteria include Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

The results indicate that the system effectively meets user requirements and performs efficiently across all ISO 25010 quality characteristics. The system's features support real-time processes and ensure efficient and reliable performance.

Table 4. *System Evaluation: Functional Suitability*

FUNCTIONAL SUITABILITY	AVERAGE MEAN	DESCRIPTIVE INTERPRETATION
Functional Completeness	4.79	Very Satisfactory
Functional Correctness	4.82	Very Satisfactory
Functional Appropriateness	4.81	Very Satisfactory
MEAN	4.81	Very Satisfactory

Table 4 presents the results of the Functional Suitability evaluation based on the responses of 110 respondents. The system obtained an average mean score of 4.81, which is interpreted as Very Satisfactory.

This indicates that the system effectively performs its intended functions and meets user requirements. The respondents observed that the system provides accurate and appropriate outputs, demonstrating strong functional performance

Table 5. System Evaluation: Performance Efficiency

PERFORMANCE EFFICIENCY	AVERAGE MEAN	DESCRIPTIVE INTERPRETATION
Time Behavior	4.74	Very Satisfactory
Resource Utilization	4.71	Very Satisfactory
Capacity	4.7	Very Satisfactory
MEAN	4.72	Very Satisfactory

Table 5 presents the results of the Performance Efficiency evaluation based on the responses of 110 respondents. The respondents consisted of 99 residents, 10 Information Technology (IT) experts/professionals, and 1 system administrator.

The system obtained an average mean score of 4.72, indicating that its overall performance was assessed as Very Satisfactory. All respondents observed that the system performed efficiently in terms of performance efficiency.

Table 6. System Evaluation: Compatibility

COMPATIBILITY	AVERAGE MEAN	DESCRIPTIVE INTERPRETATION
Co-existence	4.58	Satisfactory
Interoperability	4.64	Very Satisfactory
MEAN	4.61	Satisfactory

Table 6 presents the results of the Compatibility evaluation based on the responses of 110 respondents. The respondents 99 Residents of Palihan, Hermosa Bataan along with the Barangay Admin and Information Technology (IT) professionals.

The system obtained an average mean score of 4.61, indicating that its overall performance was assessed as Very Satisfactory. All respondents observed that the

system demonstrated good compatibility, operating effectively across different devices and environments while maintaining consistent performance and functionality for both users and administrators.

Table 7. System Evaluation: Usability

USABILITY	AVERAGE MEAN	DESCRIPTIVE INTERPRETATION
Appropriateness	4.74	Very Satisfactory
Recognizability		
Learnability	4.75	Very Satisfactory
Operability	4.76	Very Satisfactory
User Error Protection	4.53	Satisfactory
User Interface Aesthetics	4.74	Very Satisfactory
Accessibility	4.74	Very Satisfactory
MEAN	4.71	Very Satisfactory

Table 7 presents the results of the Usability evaluation based on the responses of 110 respondents. The respondents consisted of 99 residents, 10 Information Technology (IT) experts/professionals, and 1 system administrator.

The system obtained an average mean score of 4.71, indicating that its overall performance was assessed as Very Satisfactory. All respondents observed that the system exhibited high usability, as it is user-friendly, easy to navigate, and allows users to efficiently monitor product expiration and manage store inventory without difficulty.

Table 8. System Evaluation: Reliability

RELIABILITY	AVERAGE MEAN	DESCRIPTIVE INTERPRETATION
Maturity	4.79	Very Satisfactory
Availability	4.81	Very Satisfactory
Fault Tolerance	4.63	Very Satisfactory
Recoverability	4.6	Satisfactory
MEAN	4.71	Very Satisfactory

Table 8 presents the results of the Reliability evaluation based on the responses of 110 respondents. The respondents consisted of 99 residents, 10 Information Technology (IT) experts/professionals, and 1 system administrator.

The system obtained an average mean score of 4.71, indicating that its overall performance was assessed as Very Satisfactory. All respondents observed that the system demonstrated high reliability, consistently performing its functions accurately in tracking product expiration and maintaining stable operation during store monitoring processes.

Table 9. System Evaluation: Security

SECURITY	AVERAGE MEAN	DESCRIPTIVE INTERPRETATION
Confidentiality	4.7	Very Satisfactory
Integrity	4.67	Very Satisfactory
MEAN	4.69	Very Satisfactory

Table 9 presents the results of the Security evaluation based on the responses of 110 respondents. The respondents consisted of 99 residents, 10 Information Technology (IT) experts/professionals, and 1 system administrator.

The system obtained an average mean score of 4.69, indicating that its overall performance was assessed as Very Satisfactory. All respondents observed that the system ensured strong security by protecting store data and maintaining secure access through proper authentication and data handling within the application

Table 10. *System Evaluation: Maintainability*

MAINTAINABILITY	AVERAGE MEAN	DESCRIPTIVE INTERPRETATION
Modularity	4.74	Very Satisfactory
Reusability	4.61	Satisfactory
Analyzability	4.6	Satisfactory
Modifiability	4.7	Very Satisfactory
Testability	4.7	Very Satisfactory
MEAN	4.67	Very Satisfactory

Table 10 presents the results of the Maintainability evaluation based on the responses of 110 respondents. The respondents consisted of 99 residents, 10 Information Technology (IT) experts/professionals, and 1 system administrator.

The system obtained an average mean score of 4.67, indicating that its overall performance was assessed as Very Satisfactory. All respondents observed that the system demonstrated good maintainability, allowing easy updates, modifications, and efficient management of product records and inventory data.

Table 11. System Evaluation: Portability

PORTABILITY	AVERAGE MEAN	DESCRIPTIVE INTERPRETATION
Adaptability	4.69	Very Satisfactory
Installability	4.74	Very Satisfactory
Replaceability	4.75	Very Satisfactory
MEAN	4.73	Very Satisfactory

Table 11 presents the results of the Portability evaluation based on the responses of 110 respondents. The respondents consisted of 99 residents, 10 Information Technology (IT) experts/professionals, and 1 system administrator.

The system obtained an average mean score of 4.73, indicating that its overall performance was assessed as Very Satisfactory. All respondents observed that the system demonstrated high portability, as it can be effectively used across different devices while maintaining its functionality and performance.

CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATION

Summary

This capstone project focused on the design and development of a Mobile-Based Product Expiration Monitoring Application for Sari-Sari Stores in Palihan, Hermosa, Bataan. The primary goal of the study was to provide sari-sari store owners with a practical and efficient Mobile Application for monitoring product expiration dates, reducing food waste, minimizing financial losses, and ensuring consumer safety. The study addressed the common challenges encountered by sari-sari store owners, particularly the reliance on manual record-keeping using notebooks. These traditional methods often result in missed expiration dates, inaccurate inventory records, and the unintended sale of expired products. To resolve these issues, the researchers developed a mobile application that allows store owners to input product details, track expiration dates, receive timely notifications, and generate reports for better inventory management.

The research employed a mixed-methods approach, combining qualitative and quantitative techniques. Interviews and surveys were conducted among selected sari-sari store owners and clients in Palihan, Hermosa, Bataan, to gather user requirements and evaluate system performance. The Rapid Application Development (RAD) methodology guided the system development, enabling rapid prototyping, iterative testing, and continuous user feedback.

The application was developed using Android Studio, with Java and Kotlin as programming languages, XML for interface design, Firebase Firestore for database management, and cloud storage for secure data backup. The system features included

product registration, expiration tracking, notification alerts, inventory monitoring, report generation, and Barcode-based.

System evaluation was conducted using the ISO/IEC 25010 Software Quality Model, focusing on Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. The results indicated that the system effectively met the requirements of sari-sari store owners and provided a reliable, user-friendly, and efficient tool for managing product expiration.

Conclusion

Based on the results of the study, the researchers conclude that the Mobile-Based Product Expiration Monitoring Application successfully achieved its objectives. The system provided sari-sari store owners in Palihan, Hermosa, Bataan, with an effective solution to monitor product expiration dates, reduce product waste, and prevent the sale of expired goods.

Recommendations

Based on the findings, evaluation results, and observed limitations of the study, several recommendations are proposed to further improve the system. The integration of SMS-based notification support is highly recommended to enhance system reliability, particularly in areas with limited or unstable internet connectivity. This will ensure that store owners continue to receive important alerts regarding product expiration even without access to mobile data or Wi-Fi.

The following are the recommendation for this day study:

Enable offline data entry with automatic synchronization to ensure continuous recording and accurate updates once internet access becomes available.

Strengthen system security by implementing improved authentication, encrypted data storage, and enhanced database protection.

Enhance barcode scanning to support a wider range of product codes for faster and more accurate product entry.

Provide user training, manuals, and tutorials to improve system usability and encourage adoption.

Future enhancements may include sales tracking, supplier management, and automated restocking to expand system functionality.

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APPENDIX A

(INTERVIEW TRANSCRIPT)

Figure 47. *Interview Transcript*

Interview Transcript

Question 1:

Gaano na po katagal ang inyong sari-sari store?

Question 2:

Ilang produkto ang karaniwang binebenta sa inyong store?

Question 3:

Kayo po ba ang nagma-manage ng inventory ng store, o may tumutulong sa inyo?

Question 4:

Paano po ninyo mino-monitor ang stocks ng inyong tindahan?

Question 5:

Gaano kadalas po kayo nagbibilang ng stocks?

Question 6:

Nakakaranas po ba kayo ng maling bilang ng stocks?

Question 7:

May pagkakataon po ba na nagkakaroon ng expired products sa tindahan?

Question 8:

Nahihirapan po ba kayo tandaan lahat ng stocks?

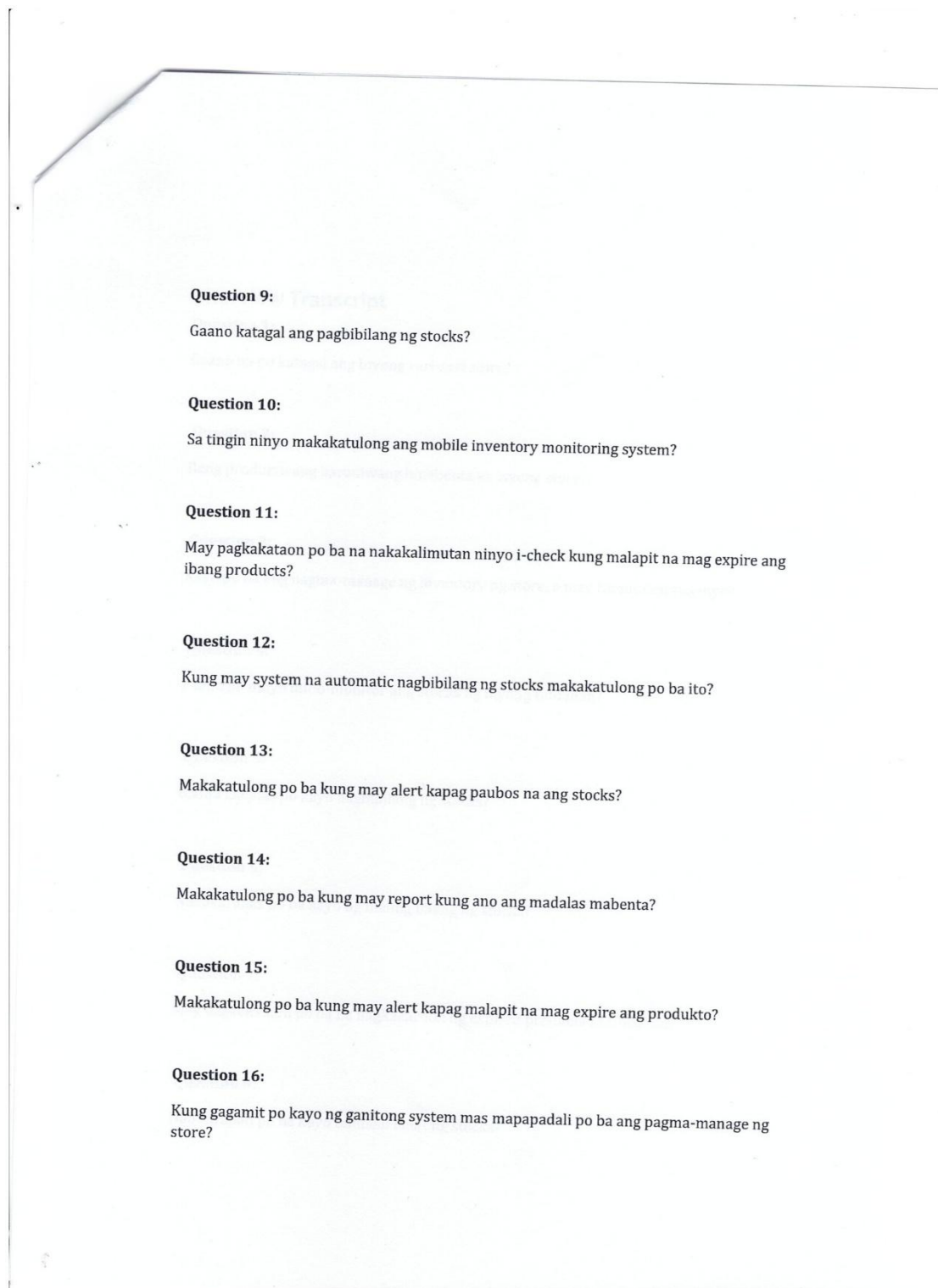
Figure 48. Interview Transcript

Figure 47 and 48 present the interview transcripts used to gather participants' responses, providing insights into the system's usability, functionality, and overall performance.

APPENDIX B
(EVALUATION TOOL)

Figure 49. *Evaluation Tool*

Evaluation Tool for User Acceptability
(Adapted from ISO 25010)
“MoniApp - Inventory Monitoring System for Sari-Sari Store”

Respondent's Profile:

Name: _____

Position / Profession: _____

Years of Service: _____ years

QUESTIONNAIRE

Please indicate a check mark (✓) under the column that best describes your responses for each item about the “Title”

Please use the rating below:

5 – Very Satisfactory
4 – Satisfactory
3 – Average
2 – Poor
1 – Very Poor

Figure 50. Evaluation Tool

Please start answering here:

Criteria	Ratings				
	5	4	3	2	1
FUNCTIONAL SUITABILITY					
Functional Completeness - Functionality covers the required tasks					
Functional Correctness - System provides the appropriate results with accuracy					
Functional Appropriateness – System functions accomplishes the desired tasks and objectives					
PERFORMANCE EFFICIENCY					
Time Behavior - Response and processing time is at acceptable rate when performing specific functionality					
Resource Utilization - Amounts and types of resources used by the system, when performing its functions, meet requirements of the user					
Capacity – Limitations like internet speed, size of database and processing time do not affect the overall performance of the system					
COMPATIBILITY					
Co-existence – The system works well with different platforms like different operating system and web browser					
Interoperability – System works well when exchanging information with different units of the organization					
USABILITY					
Appropriateness Recognizability – Manuals, tutorials, demonstration and other documentation for the use of the system are appropriately provided.					
Learnability – Use of the system is easily learned by the intended users					
Operability – Functions of system are designed to be easily adaptable for users					
User Error Protection - System protects users against making errors					

Figure 51. Evaluation Tool

User Interface Aesthetics – User interface like its colors and icons enables pleasing and satisfying interaction for the users					
Accessibility – System can be easily accessed by intended users either by internet or intranet.					
Criteria	Ratings				
	5	4	3	2	1
RELIABILITY					
Maturity – System component meet the needs for reliability under normal operation					
Availability – System component is operational and accessible when required for use					
Fault Tolerance – System component operates as intended despite the presence of hardware or software faults					
Recoverability – In the event of an interruption or a failure, the system can recover the data directly affected and re-establish its normal state					
SECURITY					
Confidentiality – System ensures that data are accessible only to those authorized users					
Integrity – System prevents unauthorized access to, or modification of, computer programs or data					
MAINTAINABILITY					
Modularity – System failure of one component has minimal impact to another parts of the computer program					
Reusability – Data from another parts of the system are easily shared with other units or entities					
Analyzability – Errors or failures of the system is easily diagnosed and mechanism to determine cause of failures are easily identified.					
Modifiability – System can be effectively and efficiently modified without introducing defects or degrading existing quality					
Testability – Effectiveness and efficiency with which test criteria can be established for the system and tests can be performed to determine whether those criteria have been met					

Figure 52. Evaluation Tool

Criteria	Ratings				
	5	4	3	2	1
PORTABILITY					
Adaptability – System can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environments					
Installability – Effectiveness and efficiency with which the system can be successfully installed and/or uninstalled in a specified environment					
Replaceability – System can replace another software for the same purpose in the same environment					

Signature over printed name of Respondent

Date

Figure 49 to 52 include the criteria and rating scales used for assessing the system's usability, functionality, and overall performance.

APPENDIX C
(DEPLOYMENT CERTIFICATE)

Figure 53. Deployment Certificate



The image shows a deployment certificate from the Office of the Punong Barangay, Barangay Palihan, Hermosa, Bataan. The certificate is dated April 21, 2026, and certifies the deployment of five students from Eastwood Professional College of Science and Technology – Balanga. The students listed are Darelle Boragay, Ralph John Santos, Nathaniel Villeza, Dave Emmanuel Dabu, and Lester Alex Alindongan. The certificate is signed by Lambertio E. Suelto, Punong Barangay. It also includes the vision and mission statements of Barangay Palihan.

 Republic of the Philippines
Province of Bataan
Municipality of Hermosa
BARANGAY PALIHAN
R.T. Estanislao Street, Palihan, Hermosa, Bataan



OFFICE OF THE PUNONG BARANGAY
BARANGAY CERTIFICATION

TO WHOM IT MAY CONCERN:

This is to certify that the following students from Eastwood Professional College of Science and Technology – Balanga:

- Darelle Boragay
- Ralph John Santos
- Nathaniel Villeza
- Dave Emmanuel Dabu
- Lester Alex Alindongan
- Algie Dorog

were officially deployed in Barangay Palihan, Hermosa, Bataan on April 21, 2026, for the conduct and implementation of their project entitled:

“Mobile-Based Product Expiration Monitoring Application for Sari-Sari Stores in Palihan, Hermosa, Bataan.”

This certification is issued upon request of the above-named individuals for whatever legal purpose it may serve.

Issued this 23rd day of APRIL, 2026 at the office of the Punong Barangay, Barangay Palihan, Hermosa, Bataan.

Certified by: 
LAMBERTIO E. SUELTO
PUNONG BARANGAY

VISION;
Barangay Palihan, is the center of progress and development, a hub for industrial and economic front, where empowered and God loving constituents enjoy a responsive social service in a safe, peaceful and environmentally sound community.

MISSION;
To provide peaceful, progressive environment, equal service and fostering cooperation from Palihan Barangay Council, practicing Barangay good governance.

This certification is valid for one month only from the date of issuance

“SAMA-SAMA SA ASENSO, PALIHENYO”

Figure 53 shows the Deployment Certificate, which serves as official documentation confirming that the system has been successfully installed, implemented, and is ready for use.

APPENDIX D

(ROUTING FORM)

Figure 54. Routing Form

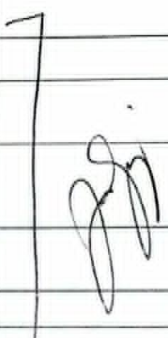


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ROUTING FORM 2

Capstone Project / Thesis Writing / Design Project Final Defense

Name: Cesar M. Mindogan
 Section: BSIT 405
 Title: Mobile Based Product Expiration Monitoring Application for Sari-Sari Store in Palihan Hermosa Bataan

PROCESS	OFFICE/PERSON-IN-CHARGE	NAME AND SIGNATURE
1. Orientation for Final Defense with accomplished Routing Form 1	Subject Instructor / Adviser	
2. Submission of Data for Statistical Analysis	School Statistician	
3. Submission of manuscript for proofreading and language editing		
4. Submission of manuscript to the Adviser for evaluation and endorsement for Final Defense (Soft Copy)	Adviser	
5. Submission of manuscript to Final Defense Panels (1 week before the date of defense)	Head Panel	
	Panel 1	
	Panel 2	
6. Defense Fee payment	Cashier	
7. Endorsement for Final Defense	Registrar	
After Final Defense		
8. Submission/Acceptance of Revised Manuscript to Defense Panel	Head Panel	
	Panel 1	
	Panel 2	
9. Acceptance of Revised Project / Prototype	Head Panel	
	Panel 1	
	Panel 2	
10. Submission of accomplished manuscript for Proofreading and Plagiarism Checking	School Proofreader	
11. Submission of Hard Bound Manuscript to the Department Coordinator / Dean (2 copies)	Department Coordinator / Dean	

Student Signature: _____

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Figure 55. Routing Form



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ROUTING FORM 2 *We Educate, Develop and Inspire*

Capstone Project / Thesis Writing / Design Project Final Defense

Name: Dave Emmanuel P. Dato
 Section: BSIT-405
 Title: Mobile-Based Product Expiration Monitoring Application for Sari-Sari Store in Palihan, Hermosa

PROCESS	OFFICE/PERSON-IN-CHARGE	NAME AND SIGNATURE
1. Orientation for Final Defense with accomplished Routing Form 1	Subject Instructor / Adviser	
2. Submission of Data for Statistical Analysis	School Statistician	
3. Submission of manuscript for proofreading and language editing		
4. Submission of manuscript to the Adviser for evaluation and endorsement for Final Defense (Soft Copy)	Adviser	
5. Submission of manuscript to Final Defense Panels (1 week before the date of defense)	Head Panel	
	Panel 1	
	Panel 2	
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7. Endorsement for Final Defense	Registrar	
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9. Acceptance of Revised Project / Prototype	Head Panel	
	Panel 1	
	Panel 2	
10. Submission of accomplished manuscript for Proofreading and Plagiarism Checking	School Proofreader	
11. Submission of Hard Bound Manuscript to the Department Coordinator / Dean (2 copies)	Department Coordinator / Dean	

Student Signature: *Dave Emmanuel P. Dato*

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Figure 56. Routing Form



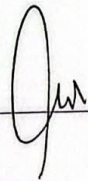
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ROUTING FORM 2

Capstone Project / Thesis Writing / Design Project Final Defense

Name: Boragay, Darrell D.
 Section: IT-405
 Title: Mobile-Based Product Expiration Monitoring Application for Sari-Sari Store in Palihan Hermosa Bat.

PROCESS	OFFICE/PERSON-IN-CHARGE	NAME AND SIGNATURE
1. Orientation for Final Defense with accomplished Routing Form 1	Subject Instructor / Adviser	
2. Submission of Data for Statistical Analysis	School Statistician	
3. Submission of manuscript for proofreading and language editing		
4. Submission of manuscript to the Adviser for evaluation and endorsement for Final Defense (Soft Copy)	Adviser	
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	Panel 1	
	Panel 2	
10. Submission of accomplished manuscript for Proofreading and Plagiarism Checking	School Proofreader	
11. Submission of Hard Bound Manuscript to the Department Coordinator / Dean (2 copies)	Department Coordinator / Dean	

Student Signature: 

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Figure 57. Routing Form

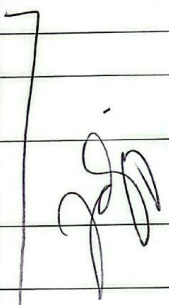


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ROUTING FORM 2

Capstone Project / Thesis Writing / Design Project Final Defense

Name: Ralph John B. Santos
 Section: IT-405
 Title: Mobile Based Product Expiration Monitoring Application for Sari Sari Stores in Palihan Hermosa Bataan

PROCESS	OFFICE/PERSON-IN-CHARGE	NAME AND SIGNATURE
1. Orientation for Final Defense with accomplished Routing Form 1	Subject Instructor / Adviser	
2. Submission of Data for Statistical Analysis	School Statistician	
3. Submission of manuscript for proofreading and language editing		
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Figure 58. Routing Form

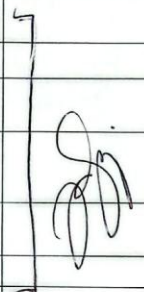


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ROUTING FORM 2

Capstone Project / Thesis Writing / Design Project Final Defense

Name: Nathaniel P. Villera
 Section: BSIT-405
 Title: Mobile-Based Product Expiration Monitoring Application for Sari-Sari Store in Palihan, Hermosa.

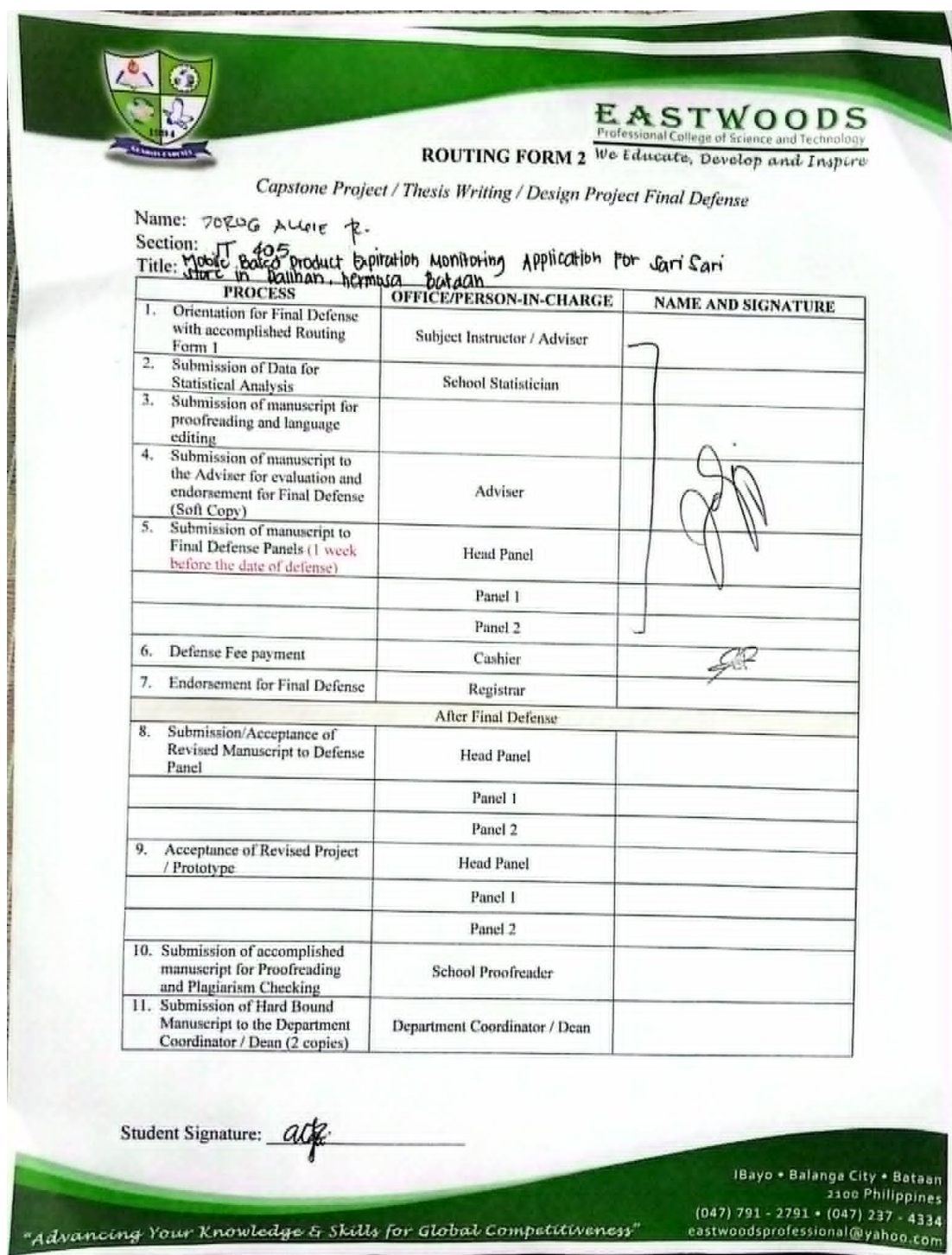
PROCESS	OFFICE/PERSON-IN-CHARGE	NAME AND SIGNATURE
1. Orientation for Final Defense with accomplished Routing Form 1	Subject Instructor / Adviser	
2. Submission of Data for Statistical Analysis	School Statistician	
3. Submission of manuscript for proofreading and language editing		
4. Submission of manuscript to the Adviser for evaluation and endorsement for Final Defense (Soft Copy)	Adviser	
5. Submission of manuscript to Final Defense Panels (1 week before the date of defense)	Head Panel	
	Panel 1	
	Panel 2	
6. Defense Fee payment	Cashier	
7. Endorsement for Final Defense	Registrar	
After Final Defense		
8. Submission/Acceptance of Revised Manuscript to Defense Panel	Head Panel	
	Panel 1	
	Panel 2	
9. Acceptance of Revised Project / Prototype	Head Panel	
	Panel 1	
	Panel 2	
10. Submission of accomplished manuscript for Proofreading and Plagiarism Checking	School Proofreader	
11. Submission of Hard Bound Manuscript to the Department Coordinator / Dean (2 copies)	Department Coordinator / Dean	

Student Signature: 

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Figure 59. Routing Form



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ROUTING FORM 2
Capstone Project / Thesis Writing / Design Project Final Defense

Name: TORUG ALUIE R.
Section: IT 405
Title: Mobile Based Product Expiration Monitoring Application for Sari Sari Store in Palihan, Hermosa, Bataan

PROCESS	OFFICE/PERSON-IN-CHARGE	NAME AND SIGNATURE
1. Orientation for Final Defense with accomplished Routing Form 1	Subject Instructor / Adviser	
2. Submission of Data for Statistical Analysis	School Statistician	
3. Submission of manuscript for proofreading and language editing		
4. Submission of manuscript to the Adviser for evaluation and endorsement for Final Defense (Soft Copy)	Adviser	
5. Submission of manuscript to Final Defense Panels (1 week before the date of defense)	Head Panel	
	Panel 1	
	Panel 2	
6. Defense Fee payment	Cashier	
7. Endorsement for Final Defense	Registrar	
After Final Defense		
8. Submission/Acceptance of Revised Manuscript to Defense Panel	Head Panel	
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9. Acceptance of Revised Project / Prototype	Head Panel	
	Panel 1	
	Panel 2	
10. Submission of accomplished manuscript for Proofreading and Plagiarism Checking	School Proofreader	
11. Submission of Hard Bound Manuscript to the Department Coordinator / Dean (2 copies)	Department Coordinator / Dean	

Student Signature: act

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(047) 791 - 2791 • (047) 237 - 4334
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Figure 54 to 59 shows the routing form used to document and track the approval process, indicating the flow of documents through designated personnel for review and authorization.